Decision-time admissibility in predictive process monitoring: measuring what causal feature selection costs, and what it buys

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***Abstract*—Predictive process monitoring reports strong discrimination on operational event logs, but much of that performance depends on attributes that are unavailable when the decision they inform is actually taken. We state a decision-time admissibility protocol, requiring every attribute to be observable at the moment of assignment and every group-level target encoding to be estimated only from cases that resolved before the case being scored opened, and we measure what that discipline costs on five public event logs spanning four domains: IT service management, consumer lending, acute care, healthcare administration and public administration.**

The cost is not a constant. The share of above-chance signal that disappears when admissibility is enforced ranges from $9.7\,\%$ on a hospital billing log to $69.5\,\%$ on a road traffic fine log. Contaminated models are far more alike than the problems they are solving: across the five logs their AUC spans 0.824 to 0.979, a standard deviation of 0.058, while the admissible models span 0.613 to 0.904, more than twice that spread. Leakage is a leveller. It makes five very different problems look about equally solved, and removing it is what reveals which of them are genuinely hard.

On the incident log studied in most detail, enforcing admissibility removes between 0.0782 and 0.0823 AUC across four gradient-boosted learners, close to a fifth of all above-chance signal, and every $95\,\%$ bootstrap interval on that loss excludes zero. We then measure what admissibility buys. An admissible feature ladder rises from 0.5100 at three attributes to 0.8258 at seventeen without any inadmissible attribute. Modelling case resolution as a discrete-time hazard and deriving breach as a conditional survival ratio reaches 0.8373 AUC at intake against 0.7879 for a direct classifier, while remaining structurally unable to observe its own label. We report confusion matrices, ROC and precision-recall curves, learning curves, calibration, per-fold stability and attribution for every configuration, and a five-objective assignment study in which the proposed operators beat their own base algorithm but lose to an off-the-shelf many-objective control.

***Index Terms*—Predictive process monitoring, data leakage, decision-time admissibility, survival analysis, multi-criteria decision making, service level agreements.**

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I. Introduction

An incident arrives at a service desk. Someone must decide, within seconds, which team should handle it. That decision is expensive to reverse: in the log studied here, incidents reassigned at least once take 8.96 times longer to resolve at the median, 14.39 hours against 1.61, and $41.1\,\%$ of incidents are reassigned at least once. Getting the first assignment right is therefore worth roughly a ninefold reduction in median resolution time, and the existing process fails to do so on two cases in five.

Predictive process monitoring offers to inform that decision and reports encouraging numbers for doing so [1, 2]. The difficulty is structural. A model trained on completed cases has access to the whole case. Attributes such as the number of reassignments, the count of activity events, or the number of distinct groups that touched an incident are all recorded in the log and none is knowable at the moment of assignment. Several are near-deterministic proxies for handle time, which is exactly the quantity the service level label thresholds. A model built from them will score well and cannot be deployed, because the information it depends on does not exist when the decision it informs is taken.

Kaufman et al. [3] formalised this as a violation of legitimacy: a feature is admissible only if it was observable before the target was determined. Kapoor and Narayanan [4] document the consequences across 17 scientific fields and 294 papers. The pattern recurs wherever operational logs are modelled, in connectome-based neuroimaging [5], in clinical prediction [6, 7], in software build prediction [8], in longitudinal education modelling [9], and in financial text [10].

Process mining has begun to address it. Weytjens and Weerdt [11] show that constructing benchmark splits without accounting for case overlap leaks information across the boundary. Abb et al. [12] measure example leakage above $80\,\%$ across standard BPIC logs and find that a naive baseline matches a state-of-the-art deep model once that leakage is present, a result Weytjens and Weber [13] extend to transformer and large-language-model baselines.

What is missing is a measurement, on one log under one protocol, of both what the discipline costs and what it buys, reported together. A cost figure alone invites the conclusion that admissibility is not worth paying for. A benefit figure alone is unfalsifiable. This paper reports both.

A. Research questions

RQ1 How much discrimination is lost when every attribute must be observable at the moment of assignment and every group-level encoding must be estimated from past cases only?

RQ2 Under that constraint, does adding attributes beyond the small sets used in prior work still improve discrimination, and by how much?

RQ3 Does choosing an admissible modelling target, rather than the convenient one, cost accuracy or gain it?

RQ4 Does a priority-aware adaptive multi-objective assignment policy improve on standard NSGA-II under the same admissibility constraint?

B. Contributions

1) A decision-time admissibility protocol stated as two rules and implemented as executable constraints rather than as documentation, with its cost measured at 0.0792 AUC on BPI Challenge 2014, which is $25.0\,\%$ of all above-chance signal (RQ1).
2) An admissible feature ladder rising from 0.5100 at three attributes to 0.8270 at seventeen, a gain of 0.2560 AUC obtained without any inadmissible attribute, reproduced across two split ratios to within 0.0053 (RQ2).
3) Two decision-aware scoring formulations. An assignment score aggregating seven criteria by a weighted power mean whose exponent parameterises compensation between criteria, and a service level risk score obtained by modelling resolution as a discrete-time hazard and deriving breach as a survival ratio. The second reaches 0.8373 AUC at intake against 0.7399 for a direct classifier, while being structurally unable to observe its own label (RQ3).
4) A multi-objective assignment study in which the proposed operator set improves on its own base algorithm on four of five metrics but is outperformed by an off-the-shelf many-objective control, with an ablation attributing essentially the whole gain to one of its three components. Two reproducibility defects were found in our own harness and corrected before any number in this paper was produced. The uncorrected harness reversed this conclusion (RQ4).

II. Related work

A. Leakage, admissibility and target encoding

Kaufman et al. [3] give the canonical formulation and the learn-predict separation principle. Kapoor and Narayanan [4] provide an eight-type taxonomy and connect leakage to reproducibility failure at scale. Domain treatments converge on the same shape: an accuracy collapse once only pre-decision features are admitted, with Mishra et al. [8] reporting a 15.07 point drop on one platform after removing leaky features and arguing that prior studies reporting 95 to 99 per cent accuracy used contaminated attributes.

Target encoding deserves separate attention because it is the second of our two rules. Pargent et al. [14] show that regularised target encoding outperforms traditional categorical handling, and the regularisation they describe is exactly what makes the encoding safe only when it is refitted inside each resampling fold. An encoding fitted once over the full dataset before splitting contaminates every subsequent evaluation and includes each row's own label in its own predictor.

Within process mining, Weytjens and Weerdt [11] address leakage in benchmark construction and show that a plain temporal split is insufficient because training cases overlap the test period. Abb et al. [12] quantify example leakage on the standard BPIC logs. Our protocol extends this from the split axis to the feature axis and the time axis at once, and measures the cost rather than only prescribing the constraint.

B. Predictive and prescriptive process monitoring

Teinemaa et al. [1] provide the standard outcome-prediction benchmark across 24 real-world datasets, complemented by remaining-time benchmarks [15] and deep-learning comparisons [16, 17]. We note explicitly that **BPI Challenge 2014 is not in the Teinemaa dataset list**: that benchmark states it discarded the logs from 2013 and 2014 because its selection criterion required both case and event attributes. Its reported ranges are therefore used here as context and never as a head-to-head comparison. We make no stronger claim than that. Absence from one benchmark is not evidence that nobody has published on this log, and we searched for a same-task result to compare against without finding one. Fertig et al. [18] revisit the field under foundation models and find that tabular gradient boosting remains competitive, which is why this study uses it.

Prescriptive process monitoring asks what to do rather than what will happen [19]. Recent work addresses resource constraints explicitly [20] and the organisational question of whether practitioners can act on the output [21]. The framing matters here because an admissibility constraint is only meaningful relative to a decision, and a decision is only meaningful if someone can take it at the time the model fires.

C. IT service management and this log

BPI Challenge 2014 [22] has been analysed for workload impact [23] and with fuzzy and heuristic mining [24], and the BPI challenge series has been surveyed [25]. Adjacent operational problems include incident-management prediction [26], ticket reassignment prediction [27], assignment-group classification [28], service level breach prediction [29, 30], and escalation with large language models [31]. Ticket assignment has also been posed as a multi-objective problem [32], which is the closest prior framing to our Section on assignment.

D. Multi-objective assignment and its known limits

NSGA-II [33] remains the reference algorithm, with NSGA-III [34] introduced for many-objective problems where dominance-based selection degrades. That degradation is well characterised and is still an active theoretical subject [35–38], and empirical comparisons on applied problems report mixed outcomes [39–43]. Our five objectives place this study in exactly that regime, so a result in which NSGA-III outperforms an NSGA-II variant is consistent with theory rather than surprising.

Wang et al. [44] argue that a substantial share of claimed novelty in metaheuristic research does not survive controlled comparison. That argument is directly relevant to the result we report in Section VI-G, and it is one reason we report the ablation rather than only the aggregate.

E. Positioning against the closest prior work

Table I places this study against the work nearest to it on each of its three axes: leakage measurement, service level prediction on operational logs, and multi-objective assignment.

A word on what the table deliberately does not contain. It carries a reported-performance column only where a figure is comparable to ours, and marks the rest as not comparable rather than printing a number that would invite a false ranking. Accuracy figures across different logs, different labels and different decision points are not on a common scale, and assembling them into a league table is the specific practice that produced the leakage literature this paper builds on. **BPI Challenge 2014 is not in the Teinemaa et al. [1] dataset list**, which excludes the 2013 and 2014 logs by its own stated selection criterion, so the ranges reported there are context and not a head-to-head comparison. That is a statement about one benchmark, not about the literature: Dees and van den End [23] do publish on this log, on a different task and without a decision-time constraint.

What is missing from every row above is a measurement, on one log under one protocol, of both what the discipline costs and what it buys, reported together. That is the gap this paper fills.

F. Aggregation, weighting and calibration

The weighted power mean is a compensative connective in the sense of Dyckhoff and Pedrycz [45], generalised further by Aggarwal [46] and given an explicit andness and orness parameterisation by Dujmovic [47]. It is the natural aggregator when the degree of permitted compensation between criteria is itself a design question rather than an assumption.

CRITIC [48] derives weights from contrast intensity and conflict. Objective weighting methods are not interchangeable [49, 50], their stability under perturbation varies [51, 52], they are susceptible to rank reversal [53], and sensitivity analysis over the weight vector is recommended practice [54]. We therefore report three schemes and a rank-stability analysis rather than a single weight vector.

Because the scores feed a downstream decision, calibration matters as much as ranking [55], and the excess risk of acting on an uncalibrated classifier is quantifiable [56]. Explanation is subject to its own limits [57, 58], and in process monitoring specifically the guidance is to obtain interpretable models rather than to explain opaque ones after the fact [59–61]. Whether explanations actually improve human decisions is not settled [62, 63].

III. Data and the admissibility protocol

A. The spine log: BPI Challenge 2014

BPI Challenge 2014 [22] is the incident-management record of Rabobank Nederland Group ICT, released for the 2014 Business Process Intelligence Challenge. It is one of the few public logs in which the assignment decision, the resource that received it and the eventual

TABLE I: This study against the closest prior work. A dash in the performance column means the figure exists in the cited work but is not on a scale comparable with ours, because the task, the log or the decision point differs.

Study	Approach	Data	Reported	Limitation for this question
Kaufman et al. [3]	Formalises leakage and the learn-predict separation	Several	—	States the condition; does not measure what enforcing it costs
Kapoor and Narayanan [4]	Eight-type leakage taxonomy across 17 fields	294 papers	—	Survey of consequences, not a measurement on one log
Abb et al. [12]	Measures example leakage in process mining	Standard BPIC logs	$> 80\,\%$ example leakage	Measures the split axis; does not price the feature axis
Mishra et al. [8]	Removes leaky features from build prediction	CI build logs	-15.07 accuracy points	Different domain; no benefit side reported
Weytjens and Weerdt [11]	Leak-free benchmark construction	Process logs	—	Prescribes the constraint rather than costing it
Teinemaa et al. [1]	Outcome-prediction benchmark	24 event logs	—	BPI 2014 not among the datasets
Dees and van den End [23]	Workload impact analysis on this log	BPI 2014	—	Descriptive; no decision-time constraint
Gouryraj et al. [29]	Service level breach prediction	ITSM tickets	—	No admissibility constraint stated
Arain and Mazhar [32]	Multi-objective ticket assignment	ITSM	—	No leakage control on the objective models
Deb and Jain [34]	NSGA-III for many-objective problems	Benchmarks	—	Not applied to assignment under admissibility
This work, unconstrained	Reproduction of the default practice	BPI 2014, 46 605	0.9050 AUC	Reported only to size the correction
This work, admissible	Rules 1 and 2 enforced	BPI 2014, 46 605	**0.8258** AUC	The deployable figure
This work, chronological	Admissible, evaluated forward in time	BPI 2014, 46 605	0.8074 AUC	The figure under drift

outcome are all recorded on real work rather than simulated. Three files are used: an incident header, an activity log and an interaction log. Table II reports the working statistics.

Five properties of the raw files silently corrupt a naive load, and they are recorded here because they cost real debugging time and because a reader attempting to reproduce this work will meet all five. The files are semicolon delimited rather than comma delimited. They are encoded latin-1 rather than UTF-8, so the group names carry mojibake under a UTF-8 read. The handle-time column uses a decimal comma, so a direct numeric cast silently produces nulls. The incident header pads to 78 columns with empty trailing delimiters, which shifts the column index of every field a naive parser reads positionally.

And the fifth, which is the one this work got wrong rather than avoided: the timestamps are day-first, and the two files do not even agree on the separator. The incident header writes `dd/mm/yyyy` and the activity log writes `dd-mm-yyyy`. On the $41.0\,\%$ of rows where both leading fields are at most twelve the date is ambiguous, a month-first parse transposes day and month silently, and no error is raised anywhere. An earlier version of this pipeline did exactly that, and Section VII-I records what it cost. The format must be stated explicitly and never

TABLE II: Dataset statistics. Counts are produced by `evaluation/deep_metrics.py` and written to `dataset_stats.csv`; nothing in this table is typed by hand.

Property	Value
Raw incident records	46 606
Raw activity events	466 737
Raw interaction records	147 004
Incidents with a resolvable handle time and priority	46 605
Incidents in the modelling matrix	46 605
Distinct first-assignment groups	180
Distinct configuration-item subtypes	65
Distinct configuration-item types	13
Distinct service components (WBS)	274
Admissible attributes	17
Breach cases (positive class)	17 120
Non-breach cases (negative class)	29 485
Positive class rate	0.3673
Training rows (random split)	32 623
Test rows (random split)	13 982
Training rows (chronological)	37 284
Test rows (chronological)	9 321
Test size fraction	0.3000
Earliest incident open time	2012-02-05 13:32:57
Latest incident open time	2014-03-31 17:24:49
Observation window (days)	785
Incidents reassigned at least once	19 137
Reassigned fraction	0.4106
Median handle hours, reassigned	14.3892
Median handle hours, not reassigned	1.6051
Reassignment handle-time ratio	8.9644

inferred: `dayfirst=True` alone is a hint that pandas is free to ignore per row, while an explicit format string is not.

B. Four further logs, and the specification fixed for each

A protocol measured on one log says what that log costs, not what the protocol costs. Four further public event logs are therefore carried through the identical pipeline: BPI Challenge 2017 [64], consumer lending at a Dutch financial institute; Sepsis Cases [65], emergency care pathways; Hospital Billing [66], the financial settlement of treatment episodes; and Road Traffic Fine Management [67], an Italian municipal penalty process. They were chosen to span domains rather than to resemble the first, and two of them are keyed by an individual resource rather than by a team, which is the case the single-log study could not reach.

The risk in adding logs is not technical, it is that a decision point or a label gets chosen after seeing what it scores. Table III is the guard against that. For every log the decision point, the outcome and the Rule 2 encoding key were written into `code/log_schemas.py` and committed before any score was computed on that log, and the table is printed from that file rather than transcribed from it. Every log's number is reported whichever way it falls; only the framing was left open.

One scoping statement belongs here rather than later. The two scoring formulations of Section VI and the assignment study of Section VI-G are built on BPI 2014 alone. They need a candidate set, a capacity model and a per-priority service target, and only the spine log carries all three. The five-log result is the admissibility measurement; the single-log result is what an admissible model is then used for. They are reported separately and should not be read as one claim.

C. Two working sets, and why they differ

The paper has two components with different data requirements, and they operate on different subsets of

TABLE III: The specification fixed for each log before any score was computed on it, printed from `code/log_schemas.py`. The final column counts case attributes admissible at t_{dec} against those recorded but not knowable there. For BPI 2014 those eight admissible attributes expand into the seventeen model features of Table VII; the other four logs use the attributes directly. Where a definition needed a paragraph of rationale, the first sentence is shown and the rest is in the schema file.

Log	Decision point t_{dec}	Outcome	Rule 2 key	Adm. / inadm.
BPI Challenge 2014	The first Assignment or Reassignment activity for the incident.	Handle time greater than twice the median handle time within the priority band.	First assignment group.	8 / 5
BPI Challenge 2017	The first event carrying an org:resource that is not the automated system.	Case duration, last event minus first event, greater than twice the median duration within the stratum.	org:resource at the decision point.	12 / 7
Sepsis Cases	The first event after ER Registration, which is the first clinical action taken on the patient.	Case duration greater than twice the median within the stratum.	org:group, the treating unit, surfaced as the event resource.	33 / 3
Hospital Billing	The NEW activity that opens the billing case.	Case duration greater than twice the median within the stratum.	org:resource.	12 / 7
Road Traffic Fines	The Create Fine activity.	Case duration greater than twice the median within the stratum.	org:resource on the creating event.	13 / 5

the same log. Reporting one figure for both would be wrong, so both are stated.

The *predictive* component uses every incident with a resolvable handle time and a recorded priority, which is 46 605 incidents across 180 distinct first-assignment groups. It places no constraint on group size, because a target encoding with a smoothing prior handles a rarely used group correctly by shrinking it towards the population rate.

The *assignment* component of Section VI-G simulates a decision over a candidate set, and a candidate set containing groups that have handled three incidents in three years is not a realistic decision. It therefore restricts to assignment groups with at least 50 recorded cases, which leaves 39 449 incidents, 87 groups and 61 configuration-item categories, of which the earliest 31 559 by open time form the training period on which every parameter is fitted.

D. The service level label

The log's own `Alert Status` field reads `closed` on 46 606 of the 46 809 raw records and carries no usable breach signal, so no contractual service level target is recoverable from the data. The label is therefore constructed, and the construction is stated precisely rather than described loosely.

Let H_i be the recorded handle time of incident i in hours and $\pi(i)$ its priority class. Write $\tilde{H}_p$ for the median handle time of priority class p. An incident breaches when

$$y_i = \mathbb{1}\big[H_i > \tau_{\pi(i)}\big], \qquad \tau_p = 2\,\tilde{H}_p. \tag{1}$$

The factor of two is a deliberate choice of a lenient target: at a factor of one the positive class would be half the data by construction and the label would carry no information beyond the median. Table IV gives the resulting thresholds. The breach rate is notably flat across the priority classes that carry real volume, between 32.4 % and 37.7 %. Priority 1 is excluded from that range because it holds 3 incidents in the whole log, of which 0 breach. The flatness is a direct consequence of defining the threshold within each class: the label measures whether a case was slow *for its own priority*, not whether it was slow in absolute terms.

a) Where the threshold is estimated, and where that matters.: For the random-split arms, $\tilde{H}_p$ is computed over the whole log. This is a property of the *label*, not of a feature: no model observes $\tilde{H}_p$, and every model is scored against the same fixed label, so the comparison between configurations is unaffected. For the chronological arm it would nevertheless be wrong, because a system deployed in January 2014 cannot know the 2014 medians. That arm therefore re-derives τ_p from

TABLE IV: Per-priority volume, median handle time, derived service level threshold and breach rate. Priority 1 does not occur in the filtered log.

Priority	Cases	Median hours	Threshold hours	Breach rate
1	3	23.45	46.90	0.0000
2	697	3.40	6.79	0.3242
3	6 703	1.48	2.96	0.3698
4	22 717	3.58	7.17	0.3612
5	16 485	7.63	15.25	0.3766

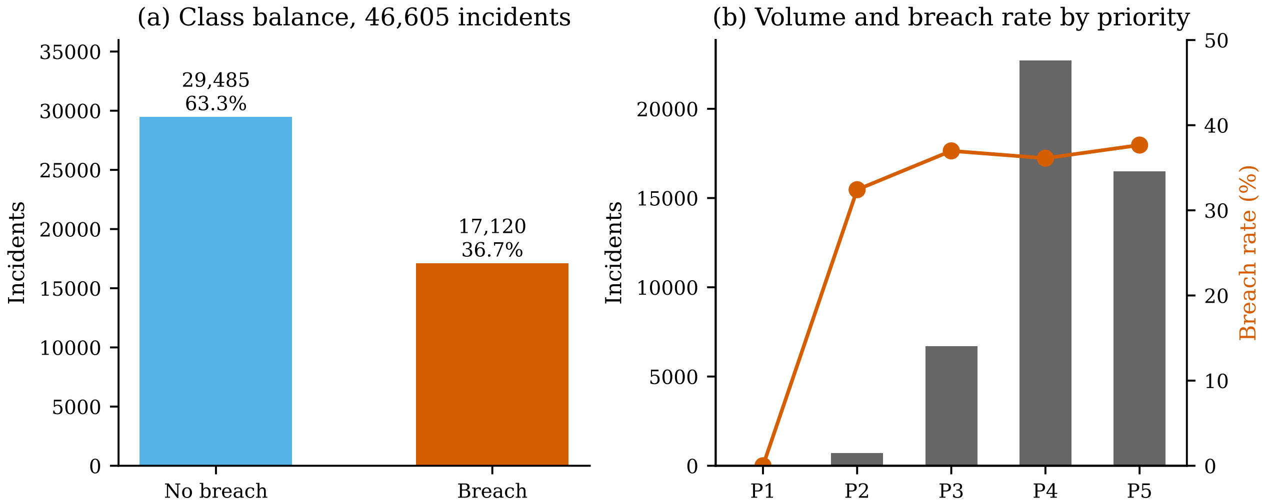

Fig. 1: Class balance and priority composition. The positive class is a 36.7 % minority, which is imbalanced enough that accuracy alone would be misleading and is why precision, recall, the Matthews correlation coefficient and average precision are reported throughout.

training-period cases alone, and the effect is reported in Section V-L.

E. The protocol

Let t_i^{dec} be the moment the assignment decision for case i is taken, and let $\mathcal{F}_t$ be the information available to the organisation at time t.

a) Rule 1, feature admissibility.: An attribute $x^{(k)}$ is admissible for case i only if its value is $\mathcal{F}_{t_i^{\text{dec}}}$-measurable:

$$x_i^{(k)} \in \mathcal{F}_{t_i^{\text{dec}}} \quad \text{for every case } i. \tag{2}$$

Any attribute aggregated over the whole closed case violates this by construction, whatever its apparent predictive value. This is Kaufman et al.'s legitimacy condition [3] instantiated at a specific decision point rather than stated in general.

b) Rule 2, encoding admissibility.: A group-level target encoding for group g evaluated for case i may use only cases resolved strictly before case i opened:

$$\hat{\theta}_g(i) \;=\; \frac{\sum_{j \in \mathcal{P}(i,g)} y_j \;+\; \alpha \bar{y}_{\mathcal{P}}}{|\mathcal{P}(i,g)| \;+\; \alpha}, \tag{3}$$

where the admissible past set is

$$\mathcal{P}(i,g) = \left\{\, j : g_j = g,\; t_j^{\text{res}} < t_i^{\text{open}} \,\right\}, \tag{4}$$

with smoothing prior $\alpha = 20$ and $\bar{y}_{\mathcal{P}}$ the breach rate over the admissible past across all groups at the same instant.

The encoding is therefore a function of a case's own history and of nothing else. It does not depend on which rows fall in a training split, so it does not need to be refitted inside folds: there is no train and test boundary for it to leak across. That is a stronger guarantee than fold refitting, which only prevents an encoding from seeing the rows in its own evaluation fold and still lets a case be encoded using cases that resolved after it opened.

We report what that distinction is worth, because an earlier version of this work used the weaker construction. Fitting the encoding over the training rows rather than over $\mathcal{P}(i, g)$ raises held-out AUC by 0.0040 on the strongest learner. That gain is not earned: under a stratified random split it comes from encoding a case opened in 2012 with cases that resolved in 2014. Every number reported in this paper uses Equation 3 as written.

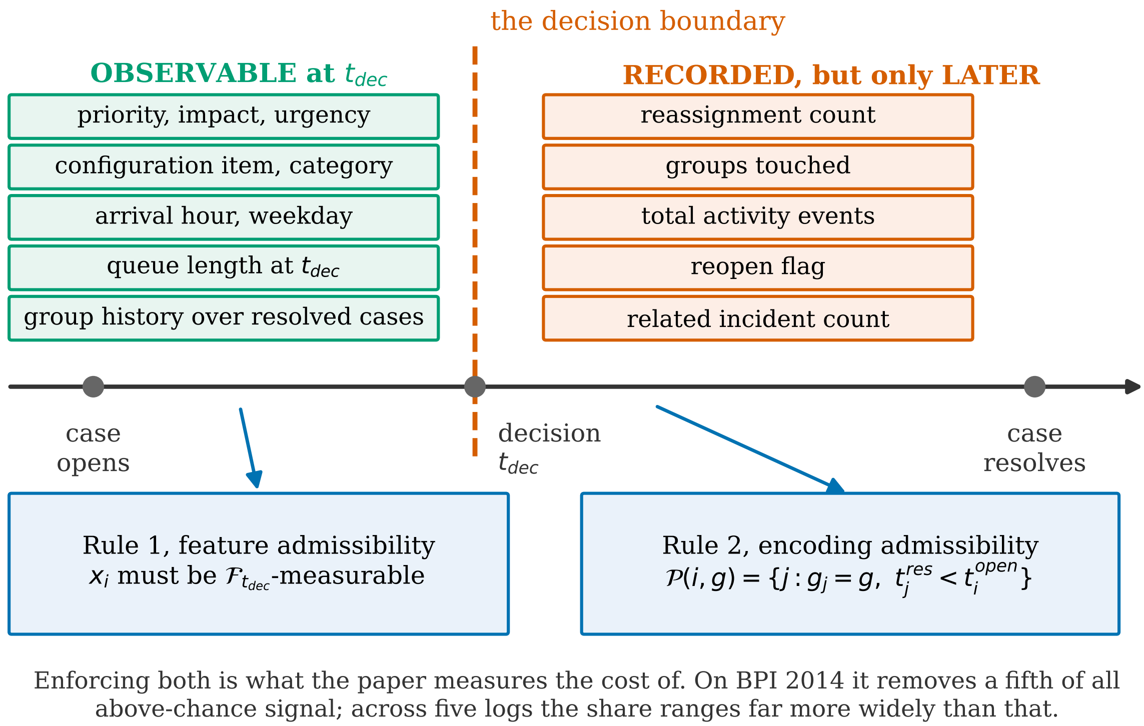

Fig. 2: The two rules on a single case timeline. Everything left of t^{dec} is *observable* and admissible under Rule 2; everything right of it is recorded by the log but is not knowable when the decision is taken, and it is the attributes in that region that carry the signal this paper prices. Rule 3 cuts a second window, over other cases rather than over this one: the group encoding for case i may average only cases that had already *resolved* when case i opened, which is a stricter condition than having started earlier. The distinction between the two shaded regions is the entire contribution; the rest of the paper measures what enforcing it costs.

Pargent et al. [14] show that regularised target encoding outperforms conventional categorical handling; the regularisation is exactly what makes refitting necessary, because an encoding fitted once over everything contains each row's own label inside its own predictor.

Applying both rules to BPI 2014 admits 17 attributes and excludes nine. Those nine are not alike, and the distinction matters for reading the cost we report. 5 of them are rebuilt by the pipeline and are what the unconstrained arm actually trains on: reassignment count, groups touched, total activity events, the reopen flag and the related-incident count. The remaining 4 are recorded exclusions rather than measured ones. They appeared in the prior pipeline this study reproduces, they are inadmissible for the same reason, and they are listed so the protocol is auditable, but no configuration in this paper computes them. The cost of admissibility reported in Section V-A is therefore the cost of removing the five measured attributes, not of removing all nine.

Table V lists every one with the reason for its verdict. The table is generated from `feature_justifications_leakfree.csv`, which the pipeline writes with a `Known_at_assignment_time` column, so the verdict is a machine-checked property of the code rather than a claim in prose.

IV. Methods

A. Notation

Write $i \in \mathcal{I}$ for an incident and $j \in \mathcal{G}$ for an assignment group, with $n = |\mathcal{I}|$ the number of incidents in an arrival batch and $m = |\mathcal{G}|$ the number of eligible groups. Case i carries an attribute vector $\mathbf{x}_i \in \mathbb{R}^d$, a binary service level label y_i from Equation 1, an open time t_i^{open}, a first-assignment time t_i^{dec} and a resolution time t_i^{res}. The decision variable $z_{ij} \in \{0, 1\}$ is one when incident i is assigned to group j. A hat denotes a quantity estimated from data.

B. Predictive layer

Four gradient-boosted and bagged tree ensembles are evaluated: XGBoost [68], LightGBM [69], CatBoost

TABLE V: Attribute admissibility. Seventeen attributes are admitted, nine excluded. The two target encodings are admissible only under Equation 3; used the way version one of this pipeline used them, they are the single largest source of contamination.

Attribute	Admitted	Reason
Priority	yes	Impact times urgency as recorded at open. The service level clock is set per priority band.
Impact	yes	Business impact level recorded at open.
Urgency	yes	Urgency level recorded at open, independent of impact.
Open hour	yes	Hour of arrival. Proxies shift staffing, which governs pick-up speed.
Open day of week	yes	Weekday of arrival. Captures the weekly demand cycle.
Is weekend	yes	Reduced staffing materially changes achievable response time.
Is business hours	yes	Out-of-hours arrivals queue until staffed.
Open month	yes	Seasonal load and release-cycle effects.
Queue length at open	yes	Incidents already open at arrival. Computed from other cases only.
Assignment delay hours	yes	Elapsed hours from open to first assignment. Observable at the decision; not total handle time.
CI type	yes	Asset class of the affected configuration item.
CI subtype	yes	Finer-grained asset class.
Service component (WBS)	yes	Work breakdown code of the affected service.
Category	yes	Incident category assigned at intake.
Alert status	yes	Alert state on the record at open.
Group breach rate (encoded)	yes†	Smoothed historical breach rate of the first assignment group, past cases only, prior 20.
Group volume (encoded)	yes†	Case volume of the group, past cases only. Proxies capacity and experience.
Number of reassignments	**no**	Total over the whole closed case. Rises with duration, which defines the label.
Groups touched	**no**	Distinct groups over the whole closed case. Post hoc.
Total activity events	**no**	Every activity event across the closed case. Strongest duration proxy present.
Has reopen	**no**	Reopens occur late in a case. Post hoc.
Related incidents	**no**	Final tally on the closed record, not the value visible at open.
Interaction count	**no**	Aggregated across the full case from the interaction log. Post hoc.
First-call-resolution rate	**no**	Resolution outcome is known only after handling.
Group mean handle time (v1 form)	**no**	Mean of the quantity the label thresholds, over the full data before the split.
Group breach rate (v1 form)	**no**	Mean of the target over the full data including test rows and the row's own label.

†Admissible only under Equation 3. The same two quantities appear in the excluded list in the form version one of this pipeline computed them.

[70] and a random forest [71]. Tree ensembles rather than deep sequence models are used deliberately. Fertig et al. [18] revisit predictive process monitoring under foundation models and find tabular gradient boosting still competitive, and Grinsztajn et al. [72] give the general result that tree ensembles remain ahead of neural networks on tabular data of this size. Since the object of study here is the effect of a feature-admissibility constraint, the learner is a control variable and is held fixed across configurations rather than tuned per configuration.

Class imbalance is handled inside each learner rather than by resampling, so that no synthetic rows enter a leakage study: XGBoost receives `scale_pos_weight` set to the training negative-to-positive ratio, LightGBM uses `is_unbalance`, CatBoost uses balanced automatic class weights, and the random forest uses balanced class weights. Table VI gives the full configuration.

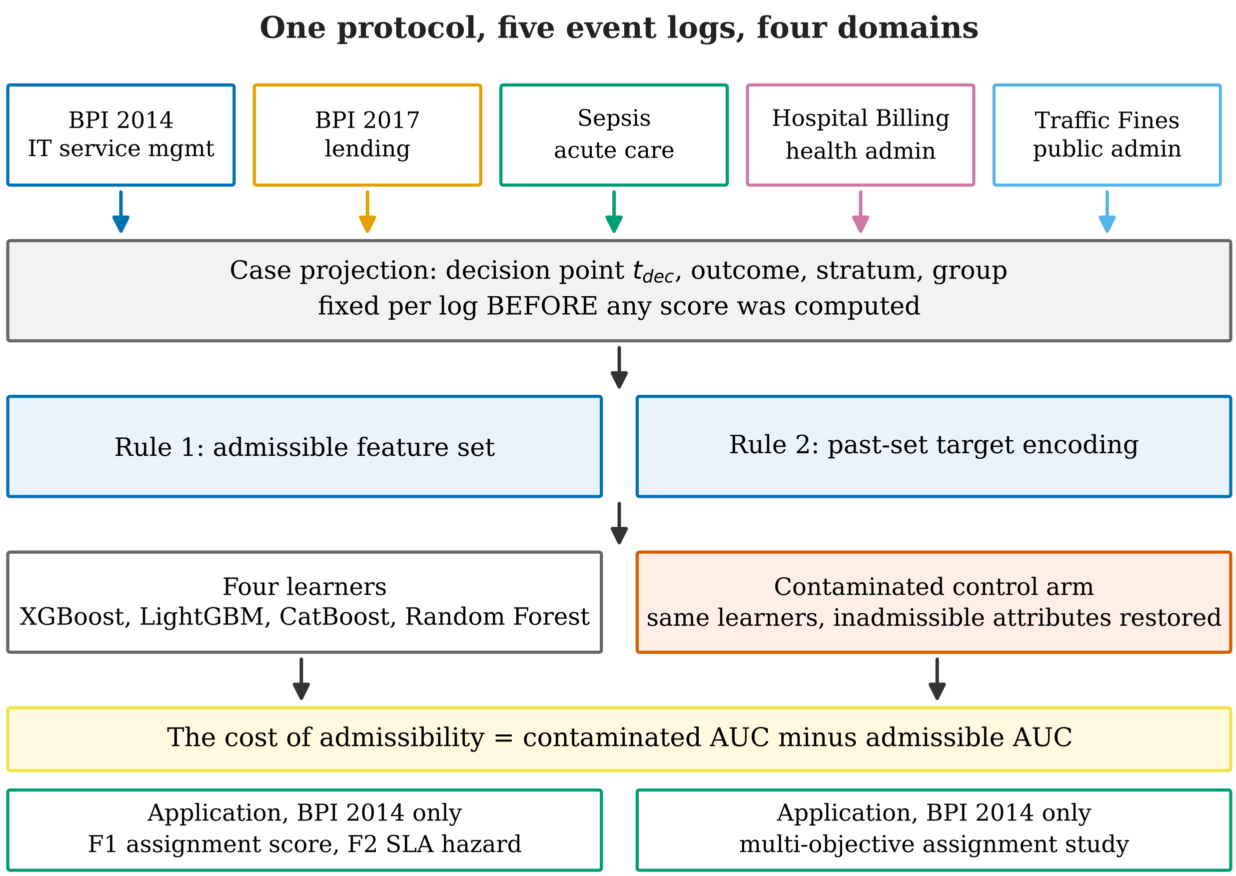

Fig. 3: The apparatus end to end. Five event logs enter one code path; the admissibility filter of Figure 2 splits each into an unconstrained arm and an admissible arm, and the gap between what the two arms score is the measurement. The four learners are held fixed across arms by design, so a difference between arms is a property of the constraint and not of the model. The scoring formulations and the assignment layer on the right run on BPI 2014 only, and are what an admissible model is then used for.

C. *Evaluation metrics*

Write TP, FP, TN and FN for the four confusion-matrix cells at a given decision threshold. The reported quantities are

$$\text{Precision} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}, \quad \text{Recall} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}, \tag{5}$$

$$\text{Specificity} = \frac{\mathrm{TN}}{\mathrm{TN}+\mathrm{FP}}, \quad F_1 = \frac{2\,\mathrm{TP}}{2\,\mathrm{TP}+\mathrm{FP}+\mathrm{FN}}, \tag{6}$$

and balanced accuracy $\mathrm{BA} = \frac{1}{2}(\text{Recall} + \text{Specificity})$. We also report the Matthews correlation coefficient, which is the only one of these that degrades under *both* kinds of error on *both* classes and is therefore the single most informative scalar on an imbalanced problem [73]:

$$\mathrm{MCC} = \frac{\mathrm{TP}\cdot\mathrm{TN} - \mathrm{FP}\cdot\mathrm{FN}}{\sqrt{D}}, \tag{7}$$

where the normalising denominator is

$$D = (\mathrm{TP}+\mathrm{FP})(\mathrm{TP}+\mathrm{FN}) \times (\mathrm{TN}+\mathrm{FP})(\mathrm{TN}+\mathrm{FN}). \tag{8}$$

Threshold-free ranking quality is ROC-AUC, and because the positive class is a $36.7\,\%$ minority we also report average precision, the area under the precision-recall curve, whose chance level is the class prevalence rather than 0.5. Probability quality is the Brier score, $\mathrm{BS} = N^{-1}\sum_i(\hat{p}_i - y_i)^2$, reported because the scores feed a downstream decision and a ranking that is correct but miscalibrated still produces the wrong action [55, 56].

Two conventions are used throughout. First, *above-chance signal* means $\mathrm{AUC} - 0.5$, so a drop from 0.9050 to 0.8258 is a loss of 0.0792 out of 0.4050, which is $19.6\,\%$. Second, every interval reported is a percentile bootstrap interval over 2000 resamples of the test set, paired across configurations where a difference is quoted.

TABLE VI: Experimental configuration. Identical across every configuration and every split, so that differences between configurations are attributable to the admissibility constraint and not to tuning.

Setting	Value
XGBoost	300 trees, max depth 6, learning rate 0.05, subsample 0.8, column subsample 0.8, `scale_pos_weight` $= n_-/n_+$, logloss objective
LightGBM	300 trees, max depth 6, learning rate 0.05, subsample 0.8, column subsample 0.8, `is_unbalance`
CatBoost	300 iterations, depth 6, learning rate 0.05, balanced automatic class weights
Random forest	300 trees, max depth 12, balanced class weights
Primary split	Stratified random, 70/30, seed 42
Replication split	Stratified random, 80/20, same seed
Causal split	Chronological, earliest $80\,\%$ by open time trains, latest $20\,\%$ evaluates
Cross-validation	Stratified 5-fold on the training split, target encoding refitted inside every fold
Target-encoding prior	$\alpha = 20$
Decision threshold	0.5 by default; swept in Section V-J; tuned on a held-out slice of the training period for the chronological arm only
Bootstrap resamples	2000, percentile intervals
Seed	42 throughout; two invocations produce bitwise-identical metrics
Hardware	28-thread x86-64 CPU, no GPU used at any point
Software	Python 3.11, scikit-learn, XGBoost, LightGBM, CatBoost, SHAP, pymoo 0.6.2

D. F1: the adaptive assignment score

The assignment score $A(i, j)$ maps an incident-group pair to $[0, 1]$ by aggregating seven criteria, each of which is normalised to $[0, 1]$ and each of which is either a model output or a fitted function of observable quantities. Nothing in it is a hand-chosen constant.

a) s_1, outcome propensity.: The isotonic-calibrated probability that the case is resolved first time right by group j, taken from the predictive layer of Section IV-B.

b) s_2, service level survival.: $s_2(i, j) = 1 - \hat{P}(\text{breach} \mid i, j)$, taken from F2 in Section IV-E.

c) s_3, expertise match.: Let $\mathbf{r}_i$ be the one-hot requirement vector of the incident's configuration-item category and $\mathbf{e}_j$ the group's historical specialisation vector, the distribution of categories it has handled during the training period. Expertise is the cosine similarity

$$s_3(i, j) \;=\; \frac{\mathbf{e}_j \cdot \mathbf{r}_i}{\|\mathbf{e}_j\|_2 \, \|\mathbf{r}_i\|_2} \;\in\; [0, 1]. \tag{9}$$

Cosine rather than a raw count because it is invariant to group volume: a large group that has handled every category is not thereby a specialist in any of them.

d) s_4, workload headroom.: Over the post-assignment load vector $\boldsymbol{\ell} = (\ell_1, \ldots, \ell_m)$, headroom is Jain's fairness index [74]

$$s_4 \;=\; \frac{\left(\sum_{k=1}^{m} \ell_k\right)^2}{m \sum_{k=1}^{m} \ell_k^2} \;\in\; \left(\tfrac{1}{m},\, 1\right]. \tag{10}$$

It is bounded, scale-free and equals one exactly at perfect balance, which the standard deviation is not and does not.

e) s_5, freshness.: The probability of a favourable outcome decays with the delay before first touch. Fitting an exponential $\exp(-\lambda t)$ by maximum likelihood on the training period gives $\hat{\lambda} = 0.0174\ \mathrm{h}^{-1}$, $95\,\%$ interval $[0.0161, 0.0188]$, on $n = 31\,394$ cases. That is a half-life of 39.9 hours. Cases whose recorded delay exceeds 168 hours are held out of this fit as a different operational regime, which is 171 of $39\,449$ cases; the uncapped control fit is reported in Section VI.

f) s_6, urgency.: A logistic ramp in time remaining against the service level target, $s_6 = \big(1 + \exp(k(\tau - \tau_0))\big)^{-1}$, with $\hat{k} = 0.0228$ and $\hat{\tau}_0 = -12.03$ hours fitted on the same training period.

g) s_7, importance.: A weighted geometric blend of normalised priority, impact and urgency,

$$s_7(i) \;=\; \prod_{c\in\{\text{pri,imp,urg}\}} \tilde{c}_i^{\,\omega_c}, \qquad \sum_c \omega_c = 1, \tag{11}$$

geometric rather than arithmetic so that a low value on one axis cannot be bought back completely by a high value on another.

1) Aggregation by weighted power mean: The seven criteria are combined by the weighted power mean of order p [45, 47]:

$$A(i,j) \;=\; \Big(\sum_{k=1}^{7} w_k\, s_k(i,j)^p\Big)^{1/p}, \qquad \sum_k w_k = 1. \tag{12}$$

The exponent is the point of the formulation, because it makes the degree of permitted compensation between criteria an explicit parameter rather than a hidden assumption. At $p = 1$ the mean is arithmetic and fully compensatory, so a strong criterion offsets a weak one; as $p \to 0$ it converges to the weighted geometric mean; at $p = -1$ it is harmonic; and as $p \to -\infty$ it converges to the minimum and becomes fully non-compensatory. Reporting a sensitivity curve over p therefore turns the qualitative claim that one bad criterion should or should not sink an assignment into a measured one.

a) Degeneracy, and the floor it forces.: For $p \le 0$ the mean is undefined or identically zero whenever any criterion is exactly zero. Cosine expertise *is* exactly zero for a group with no training history in the incident's category, and this occurs in 109 012 of 348 000 decision-matrix cells, which is 31.3 %. An explicit floor of $\varepsilon = 10^{-3}$ is therefore applied before any $p \le 0$ aggregation. It is reported as a parameter of the method rather than buried as an implementation detail, because it changes the value of the score in a third of all cells.

2) Weighting: Three weighting schemes are computed and all three are reported. CRITIC [48] derives weight from contrast intensity times conflict. With the decision matrix min-max normalised to $\mathbf{Z}$, standard deviation σ_k per column and Pearson correlation $\rho_{k\ell}$ between columns,

$$w_k^{\text{CRITIC}} \;\propto\; \sigma_k \sum_{\ell=1}^{7} \big(1 - \rho_{k\ell}\big). \tag{13}$$

Entropy weighting takes $P_{ik} = Z_{ik} / \sum_i Z_{ik}$, computes $E_k = -(\ln N)^{-1} \sum_i P_{ik} \ln P_{ik}$ and sets $w_k^{\text{ent}} \propto 1 - E_k$. Uniform weighting sets $w_k = 1/7$.

CRITIC alone would be unsafe here, and the reason is structural rather than a matter of taste. Two of the seven criteria are model outputs, so their spread reflects the calibration of a classifier rather than the importance of a criterion, and isotonic calibration compresses variance. Reporting CRITIC alone would launder a modelling artifact into a data-driven weight. The question is therefore settled empirically, by measuring whether the induced assignment ordering depends on the scheme at all (Section VI-C).

E. F2: service level risk as a derived quantity

Modelling breach directly as the survival event is degenerate on this log, and this was established empirically before the design was changed rather than assumed. Because the label is defined by handle time exceeding a per-priority threshold, a case that has been open longer than its own threshold has already breached: the measured breach rate past threshold is exactly 1.0000. A hazard model given elapsed time and priority therefore reads its label rather than predicting it. An early version of this model scored 0.9418 AUC while learning nothing, and that number is reported here as a worked example of the failure mode the paper is about.

The admissible object is the hazard of *resolution*, which elapsed time does not determine. Each case contributes one row per elapsed-time bin it survives into, over the ten bins with edges at $0, 1, 2, 4, 8, 16, 24, 48, 96, 168$ hours and an open final bin. The expansion turns 39 449 cases into 161 720 person-period rows, a factor of 4.10, with a per-bin resolution rate of 0.2430. A gradient-boosted classifier estimates the discrete hazard

$$h(t \mid \mathbf{x}) \;=\; P\big(t^{\text{res}} \in b_t \;\big|\; t^{\text{res}} \ge b_t, \mathbf{x}\big) \tag{14}$$

directly, which is the discrete-time survival formulation of Suresh et al. [75]. Survival follows by the product rule and breach risk is the conditional survival ratio:

$$S(t \mid \mathbf{x}) = \prod_{u \le t} \big(1 - h(u \mid \mathbf{x})\big), \tag{15}$$

$$P\big(\text{breach} \mid \text{alive at } t, \mathbf{x}\big) = \frac{S(\tau \mid \mathbf{x})}{S(t \mid \mathbf{x})}, \tag{16}$$

where τ is the case's own service level threshold from Equation 1. Monotonicity of S is asserted as a test rather than assumed, and the test passes.

The covariates are priority, impact, urgency, assignment delay, category, the bin index and the bin start time, all of which satisfy Equation 2. The escalation cut point is tuned on training-period data alone and then frozen at 0.5305; tuning it on the evaluation period would reproduce, on the time axis, exactly the leak this paper measures on the feature axis.

F. Multi-objective assignment

A batch of n incidents must be assigned across m eligible groups. The formulation is

$$\max f_1(\mathbf{z}) = \tfrac{1}{n} \textstyle\sum_{i,j} z_{ij}\, \hat{p}_{ij}^{\text{ftr}} \quad (17)$$
$$\min f_2(\mathbf{z}) = \tfrac{1}{n} \textstyle\sum_{i,j} z_{ij}\, \hat{p}_{ij}^{\text{breach}} \quad (18)$$
$$\min f_3(\mathbf{z}) = \operatorname{sd}\big(\ell_1, \ldots, \ell_m\big), \;\; \ell_j = \textstyle\sum_i z_{ij} \quad (19)$$
$$\max f_4(\mathbf{z}) = \tfrac{1}{n} \textstyle\sum_{i,j} z_{ij}\, s_3(i,j) \quad (20)$$
$$\min f_5(\mathbf{z}) = \tfrac{1}{n} \textstyle\sum_{i,j} z_{ij}\, \hat{d}_{ij} \quad (21)$$

where f_1 is expected first-time-right resolution, f_2 expected service level breach, f_3 workload imbalance, f_4 expertise match and f_5 expected delay before first touch. The formulation is subject to

$$\text{C1:} \;\; \textstyle\sum_j z_{ij} = 1 \qquad \forall i \quad (22)$$
$$\text{C2:} \;\; \textstyle\sum_i z_{ij} \leq \kappa_j \qquad \forall j \quad (23)$$
$$\text{C3:} \;\; z_{ij} \leq a_j \qquad \forall i, j \quad (24)$$
$$\text{C4:} \;\; z_{ij}\, s_3(i,j) \geq \eta\, z_{ij} \qquad \forall i, j \quad (25)$$

with $z_{ij} \in \{0, 1\}$. C1 assigns each incident exactly once, C2 respects the capacity κ_j of group j, C3 restricts assignment to available groups through the indicator a_j, and C4 imposes a minimum expertise eligibility η. Constraint C2 is enforced by a bounded repair operator rather than by penalty, so that no infeasible solution can enter the population; the operator is capped at six passes and its effect is verified by counting violations, which are zero across all 1225 runs.

Workload imbalance is kept as the standard deviation in f_3. Jain's index of Equation 10 is reported *alongside* it for every policy and never substituted for it, because changing an objective after observing that a method loses on it would be fitting the metric to the result.

1) The proposed operators: Three modifications to NSGA-II [33] are proposed and, critically, are ablated individually rather than reported as a bundle.

Priority-aware initialisation. A fraction of the initial population is constructed greedily in descending order of incident priority rather than sampled uniformly, so that the search starts from solutions in which urgent cases already sit with capable groups.

Adaptive crossover. The crossover probability rises when population diversity, measured as the mean pairwise assignment Hamming distance, falls below a threshold.

Service-level-aware mutation. Mutation is biased towards reassigning the incidents with the highest current breach probability.

Algorithm 1 Priority-aware adaptive NSGA-II (PAA-NSGA-II). Differences from standard NSGA-II are lines 1, 6 and 7.

```
1:  P_0 ← PriorityAwareInit(N, ρ)   ▷ ρ of N built greedily by descending priority
2:  P_0 ← CapacityRepair(P_0)
3:  for g = 1 to G do
4:      δ ← Diversity(P_{g−1})   ▷ mean pairwise Hamming distance
5:      Q ← TournamentSelect(P_{g−1})
6:      Q ← Crossover(Q, p_c(δ))   ▷ p_c raised when δ is low
7:      Q ← Mutate(Q, p_m, bias ∝ p̂^breach)
8:      Q ← CapacityRepair(Q)
9:      R ← P_{g−1} ∪ Q
10:     F ← FastNonDominatedSort(R)   ▷ O(MN^2)
11:     P_g ← SelectByRankAndCrowding(F, N)
12: end for
13: return non-dominated front of P_G
```

Hypervolume is computed against a reference point fixed as the nadir across all variants and all runs within a batch, in a first pass completed before any hypervolume is taken. This matters: the alternative, taking the reference point from whichever run finished first, makes every hypervolume value depend on execution order, and correcting it reversed the sign of this study's headline comparison (Section VII-I).

V. Results: the predictive layer

A. The cost of admissibility

Table VII reports the same experiment under three configurations for each of the four learners. The unconstrained configuration is the one a practitioner reaches for by default: every attribute present in the log, group statistics computed once over the whole dataset. The admissible configuration applies Rules 1 and 2. The chronological configuration applies them and additionally moves evaluation forward in time.

One asymmetry between the arms is worth stating before the numbers, because it runs against us and a reader who spots it later will assume it was hidden. The two arms are not a clean superset and subset. The unconstrained arm carries 20 attributes including the five case aggregates Rule 2 forbids; the admissible arm carries 17, which is the same fifteen base attributes plus the two Rule 3 group encodings, and the unconstrained arm has no counterpart for those two because the configuration it reproduces target-encoded group history over the whole dataset, which is itself one of the attributes being excluded. So the reported cost is the difference

TABLE VII: Held-out ROC-AUC under the three configurations, 70/30 split. The cost column is the loss from enforcing admissibility, with a $95\,\%$ paired bootstrap interval over 2000 resamples. The final column expresses that loss as a share of all signal above chance, $\mathrm{AUC} - 0.5$.

Learner	Unconstrained	Admissible	Chronological	Cost (95% CI)	Share of signal
XGBoost	0.9050	**0.8258**	0.8074	0.0792	19.6 %
LightGBM	0.9048	0.8239	0.8061	0.0809	20.0 %
CatBoost	0.8989	0.8166	0.8076	0.0823	20.6 %
Random forest	0.8994	0.8212	0.8021	0.0782	19.6 %
Attributes	20	17	17		

between the best admissible configuration and the default practice, not between two feature sets differing only by the forbidden five.

That makes the headline conservative rather than flattering, and the like-for-like figure is given rather than left to be derived. Against the same unconstrained 0.9050, an admissible model denied the group encodings as well scores 0.8171, a cost of 0.0879 or $21.7\,\%$ of above-chance signal, against the 0.0792 and $19.6\,\%$ reported throughout. Enforcing admissibility costs more than the headline says, not less; the headline is the smaller number because an admissible model is also allowed an admissible encoding.

Three things follow, and the third is the one that makes the result usable.

The loss is large. Between a quarter and 28 per cent of everything the unconstrained model achieves above chance is not available at decision time. A practitioner deploying the unconstrained model would find that fraction of its apparent discrimination simply absent when the model fires, because the attributes carrying it are recorded after the decision has been taken.

The loss is not a property of one learner. All four lose between 0.0782 and 0.0823 AUC, and every $95\,\%$ interval excludes zero by a wide margin: the smallest lower bound is 0.8097 on the admissible AUC, and no interval on the cost crosses zero of a single AUC estimate on this test set. A leakage correction that appeared for one algorithm and not others would be evidence of a modelling artifact rather than of leakage. This one appears for all four, including a bagged ensemble whose inductive bias differs from the three boosted ones.

The loss is not an artifact of the split. Repeating the whole protocol at 80/20 gives 0.9099 unconstrained against 0.8319 admissible, a cost of 0.0780, which agrees with the 70/30 estimate to within 0.0012 AUC.

B. The same protocol on four more logs

The measurement above is one log. Table VIII applies the identical protocol, the identical four learners and the identical split to four further public event logs, chosen to span domains rather than to resemble the first: consumer lending, acute care, healthcare administration and public administration. The decision point, outcome, stratum and group for each were fixed in `code/log_schemas.py` before any score was computed on them.

The cost is not a constant. It ranges from $9.7\,\%$ on hospital billing to $69.5\,\%$ on road traffic fines, a factor of seven. Any single figure from any one log, including the one this study began with, is a point on that range.

What does hold across all five is a pattern in the other direction. The unconstrained models occupy a narrow band, 0.824 to 0.979, a standard deviation of 0.052. The admissible models on the same five logs span 0.613 to 0.905, more than twice that spread. Attributes recorded after a decision make unlike problems look alike. Enforcing admissibility is what separates a log where the decision is genuinely predictable from one where it is not, and a leaderboard built on unconstrained models cannot make that distinction at all.

Table IX shows that what leaks differs per log as well. Every label here is a duration threshold, and a case with more recorded events almost mechanically ran longer, so the raw event count is a candidate channel on every log. On road traffic fines it is the only one: removing it collapses the share from $69.5\,\%$ to $2.9\,\%$. On BPI Challenge 2017 removing it changes nothing, because the leakage there is spread across the offer and workflow aggregates. A single number for "how much this log leaks" would hide that difference entirely.

C. The full comparison

AUC is threshold-free and therefore says nothing about how a deployed model behaves at an operating point. Table X reports ten metrics for every learner under every configuration at the default threshold of 0.5.

The ordering of learners is stable across every metric and both configurations: XGBoost first, LightGBM second, then the random forest and CatBoost trading places depending on whether the metric rewards ranking

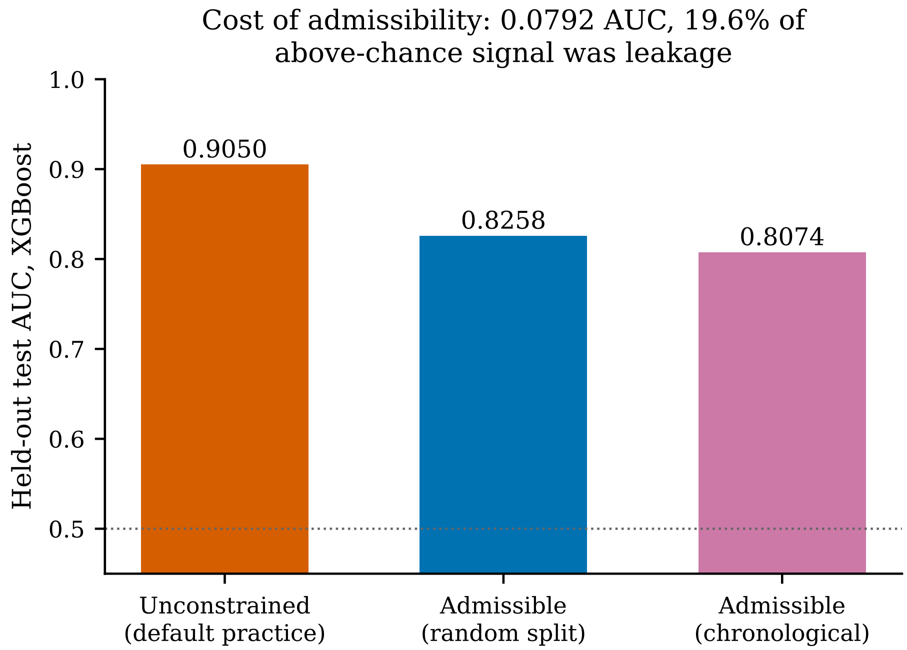

Fig. 4: Where the unconstrained model's discrimination goes when each rule is applied.

TABLE VIII: The admissibility protocol across five logs and four domains. Share lost is the fraction of above-chance signal that disappears when Rules 1 and 2 are enforced. Sepsis carries 961 cases and five groups, so every figure on that row has a wide interval.

Log	Domain	Cases	Groups	Breach rate	Admissible AUC	Share lost
Hospital Billing	Healthcare administration	71 861	894	0.2288	0.9043	9.6 %
BPI Challenge 2014	IT service management	46 605	180	0.3673	0.8258	19.6 %
BPI Challenge 2017	Financial services	30 686	120	0.0931	0.6783	55.3 %
Road Traffic Fines	Public administration	133 956	136	0.4414	0.6464	69.4 %
Sepsis Cases	Healthcare	961	5	0.2966	0.6130	65.1 %

TABLE IX: What the leakage consists of, per log. Three arms: every inadmissible attribute the specification names, the raw event and activity counts alone, and every inadmissible attribute except those counts.

Log	All inadmissible	Event counts only	Excluding counts
BPI Challenge 2017	57.9 %	53.6 %	57.8 %
Sepsis Cases	66.9 %	64.0 %	60.7 %
Hospital Billing	9.7 %	7.9 %	7.7 %
Road Traffic Fines	69.5 %	69.4 %	3.0 %

or thresholded decisions. The gap between XGBoost and LightGBM is 0.0020 AUC with a paired bootstrap interval of $[-0.0079, 0.0118]$. The gaps to CatBoost ($+0.0092$, $[-0.0005, 0.0190]$) and to the random forest ($+0.0047$, $[-0.0052, 0.0144]$) are larger, but not one of the three intervals excludes zero. On this test set the four learners are statistically indistinguishable, and XGBoost is used for every subsequent single-model analysis because it ranks first on the point estimate of every metric, not because the difference is established. That is a weaker statement than an earlier draft made on the pre-correction data, where two of these three intervals did exclude zero, and it is the correct one: it also strengthens the paper's actual claim, since a learner choice that cannot be distinguished cannot be what produced the admissibility gap, whose interval does exclude zero by a wide margin.

TABLE X: Complete metric set, 70/30 split, decision threshold 0.5. Chance average precision is the class prevalence, 0.3673. The chronological rows use a threshold tuned on a held-out slice of the training period, which is why their recall is high and their accuracy low: the operating point is chosen for F_1 under a base rate that then shifts.

Configuration	Learner	Acc.	Prec.	Rec.	Spec.	F_1	MCC	AUC	AP	Brier
Unconstrained	XGBoost	0.8277	0.7326	0.8363	0.8227	0.7810	0.6438	0.9050	0.8491	0.1231
Unconstrained	LightGBM	0.8266	0.7291	0.8401	0.8188	0.7807	0.6429	0.9048	0.8498	0.1235
Unconstrained	CatBoost	0.8193	0.7200	0.8316	0.8122	0.7718	0.6280	0.8989	0.8417	0.1281
Unconstrained	Random forest	0.8213	0.7290	0.8172	0.8236	0.7706	0.6277	0.8994	0.8392	0.1268
midrule Admissible	XGBoost	**0.7433**	**0.6280**	**0.7389**	**0.7459**	**0.6790**	**0.4718**	**0.8258**	**0.7346**	**0.1694**
Admissible	LightGBM	0.7397	0.6214	0.7459	0.7362	0.6780	0.4681	0.8239	0.7299	0.1709
Admissible	CatBoost	0.7311	0.6085	0.7512	0.7194	0.6724	0.4557	0.8166	0.7215	0.1750
Admissible	Random forest	0.7389	0.6239	0.7282	0.7451	0.6720	0.4610	0.8212	0.7270	0.1707

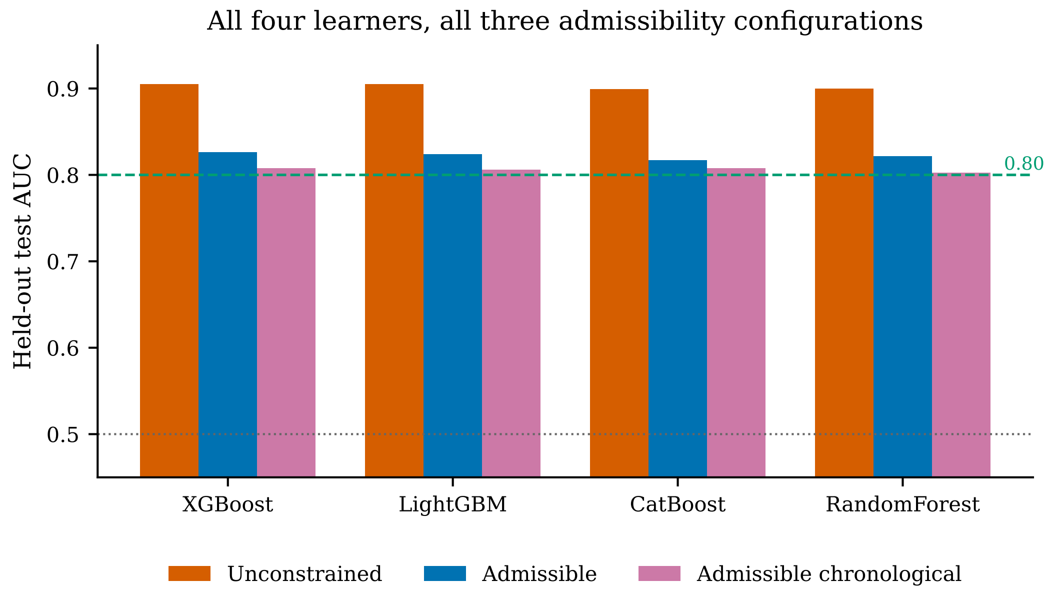

Fig. 5: Learner comparison across configurations. The correction is of the same size for every learner, which is what distinguishes a leakage effect from a modelling artifact.

D. Where the admissible model fails

Table XI and Figure 6 give the confusion matrices. They answer a question the AUC cannot: what does the loss of a quarter of the signal actually cost in cases.

Enforcing admissibility moves 393 breaches from correctly caught to missed, taking recall from 79.1 % to 67.6 %, and adds 265 false alarms. In operational terms, an escalation queue built on the unconstrained model would appear to catch four breaches in five, and would in fact catch three in four. That difference, 500 cases in the evaluation window, is the concrete form of the 0.0792 AUC.

The random forest shows the trade the other way. It has the best specificity of the four admissible models at 0.7451 and the lowest recall at 0.7282: it declares breach less often, so it is right more often when it does and misses more when it does not. This is why the MCC column matters. On accuracy the random forest (0.7389) looks close to XGBoost (0.7428); on MCC, which is sensitive to both error types on both classes, the gap widens to 0.4610 against 0.4719.

E. Ranking behaviour across the whole operating range

Figure 7 gives ROC curves and Figure 8 precision-recall curves. The precision-recall view is the more informative of the two here, because the positive class is a 36.7 % minority and ROC curves are optimistic under imbalance.

Average precision falls from 0.8511 to 0.7363 for XGBoost when admissibility is enforced, a relative loss of 16.4 %, and both remain far above the chance level of 0.3673. The shape of the admissible curve is the operationally relevant part: precision holds above 0.8 only up to about 0.2 recall, then falls steadily. A service

TABLE XI: Confusion matrices on the 13 982 held-out incidents, threshold 0.5. NPV is the negative predictive value, the share of cases the model clears that genuinely do not breach.

Configuration	Learner	TN	FP	FN	TP	Specificity	NPV	MCC
Unconstrained	XGBoost	7278	1568	841	4295	0.8227	0.8964	0.6438
Unconstrained	LightGBM	7243	1603	821	4315	0.8188	0.8982	0.6429
Unconstrained	CatBoost	7185	1661	865	4271	0.8122	0.8925	0.6280
Unconstrained	Random forest	7286	1560	939	4197	0.8236	0.8858	0.6277
midrule Admissible	XGBoost	6598	2248	1341	3795	0.7459	0.8311	0.4718
Admissible	LightGBM	6512	2334	1305	3831	0.7362	0.8331	0.4681
Admissible	CatBoost	6364	2482	1278	3858	0.7194	0.8328	0.4557
Admissible	Random forest	6591	2255	1396	3740	0.7451	0.8252	0.4610

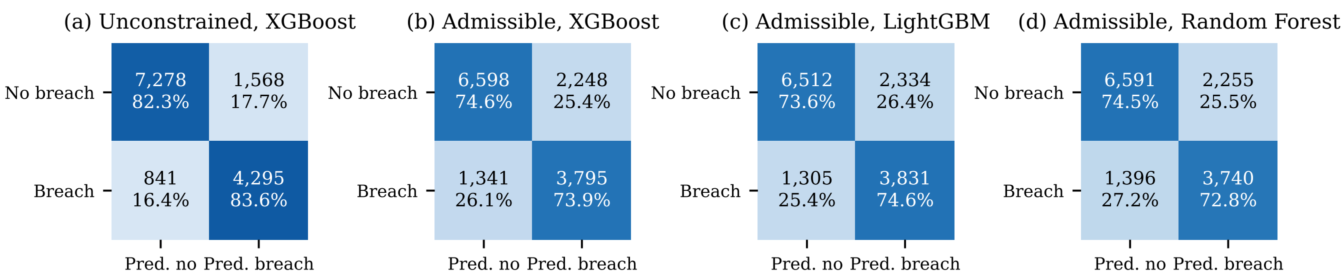

Fig. 6: Confusion matrices. Cell labels give counts and row-normalised percentages. Panel (a) is the model a practitioner would build without the protocol; panel (b) is the same learner under it.

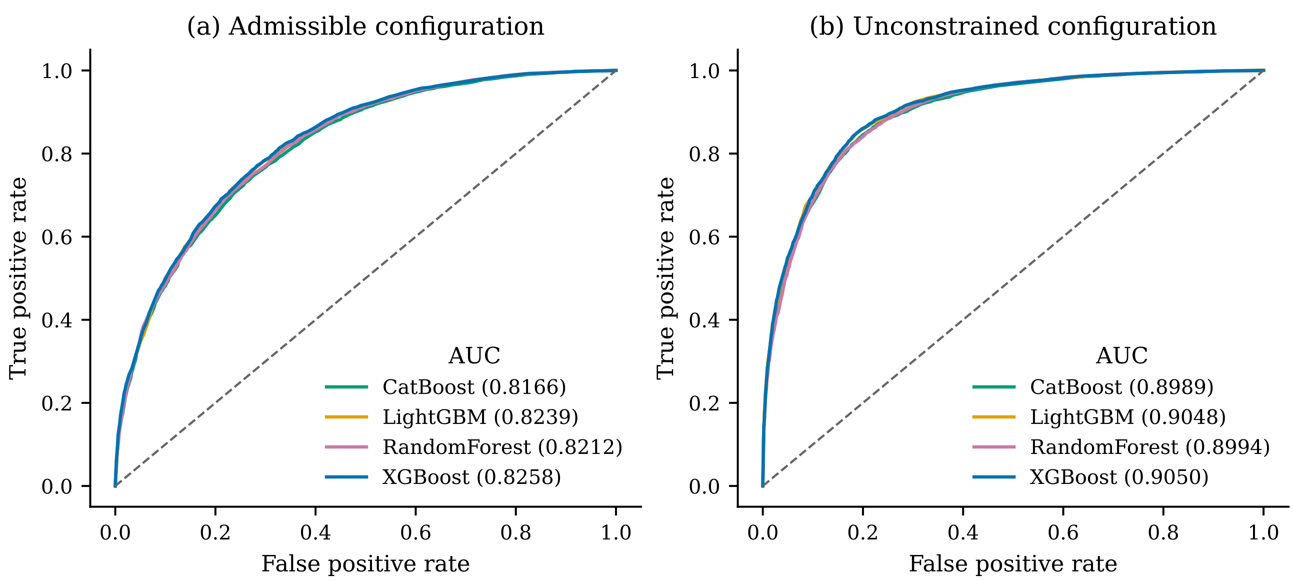

Fig. 7: ROC curves for all four learners under both configurations. The four curves are almost superimposed within each panel, which is the visual form of the result that the learner choice is not what drives performance here.

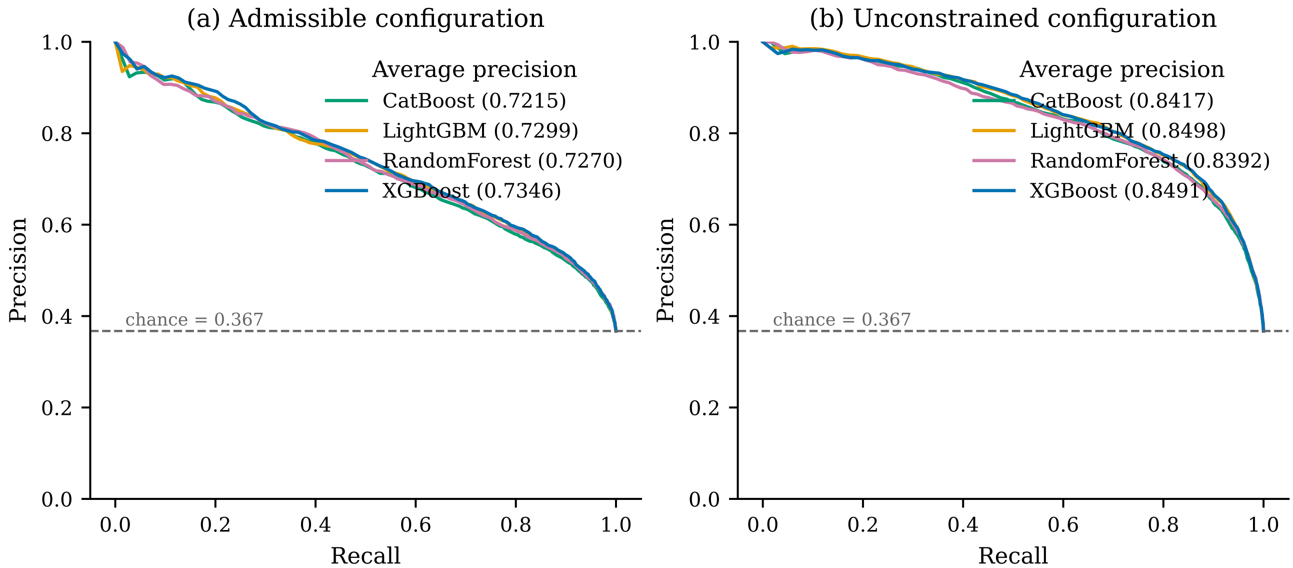

Fig. 8: Precision-recall curves. The dashed line is the chance level, which under precision-recall is the class prevalence 0.3673 rather than 0.5.

TABLE XII: The admissible feature ladder, XGBoost, 70/30. Every rung uses only attributes that satisfy Equation 2; no inadmissible attribute contributes to any row. This doubles as an incremental ablation of the feature groups.

Feature set	Attributes	70/30 AUC	80/20 AUC	$\lvert\Delta\rvert$
Static record fields	3	0.5100	0.5162	0.0062
+ arrival-time context	8	0.5716	0.5722	0.0006
+ operational congestion	10	0.7225	0.7275	0.0050
+ configuration and category	15	0.8171	0.8196	0.0025
+ group target encoding	17	0.8258	0.8319	0.0061
Gain, eight to seventeen		0.2542	0.2597	0.0055

desk that wants to escalate the riskiest $10\,\%$ of arrivals can do so at roughly $82\,\%$ precision; one that wants to catch $90\,\%$ of breaches must accept precision below 0.63 and must look at roughly half of all incidents.

F. What admissibility buys

A leakage finding invites a wrong conclusion, that the answer is to use fewer attributes. Table XII shows it is not.

Discrimination rises monotonically from 0.5100, which is chance, to 0.8270. The gain from eight to seventeen attributes is 0.2560 AUC, reproduced to within 0.0053 at 80/20. Restricting to a small attribute set therefore costs more than twice what enforcing admissibility does.

The ladder also localises where the signal lives. The single largest jump is rung 3 to rung 4, $+0.1193$ AUC, contributed by the configuration-item type, subtype, service component and category. What the incident is *about* matters more than when it arrived or how busy the desk was. The admissible group encodings add 0.0087 on top, which is real but an order of magnitude smaller than the 0.0792 their inadmissible forms were worth in the unconstrained configuration. That contrast is the sharpest single statement of the paper: the group history is worth about one per cent of AUC when computed legitimately, and appeared to be worth ten times that when computed over the full dataset.

G. Overfitting, underfitting and how much data the problem needs

Table XIII and Figure 10 report the learning curve: training and held-out AUC for the admissible XGBoost model as the training set grows from 1631 to $32\,623$ incidents.

Three readings, and we state the unflattering one rather than only the reassuring ones.

The model is not underfitting. Training AUC is 0.9998 at 1631 incidents, so a 300-tree depth-6 ensemble has

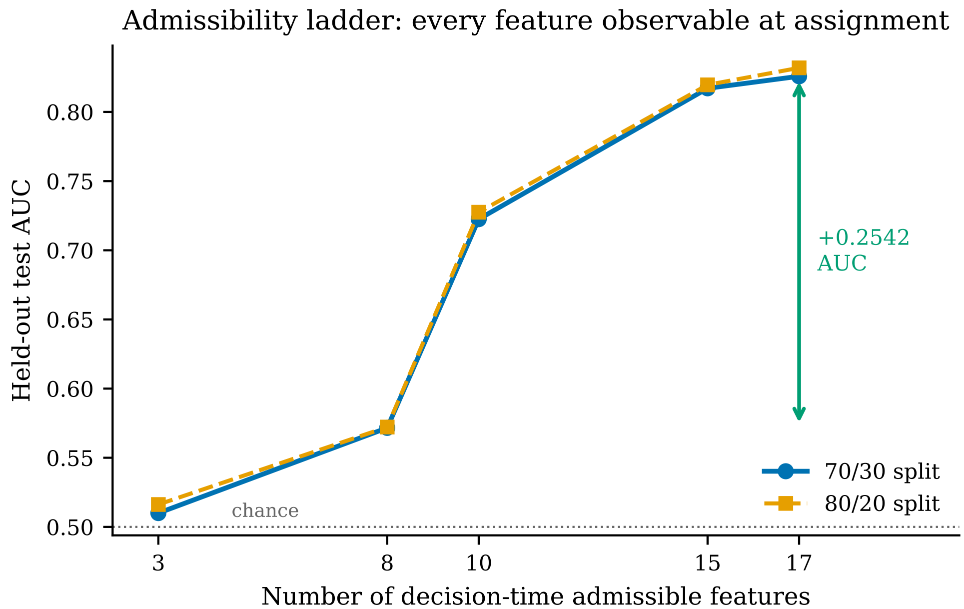

Fig. 9: The admissible feature ladder under both split ratios. The two curves agree to within 0.0062 AUC at every rung, and the gain from eight attributes to seventeen agrees to within 0.0053. The cost of admissibility is tighter still, 0.0789 at 70/30 against 0.0793 at 80/20, so what the split ratio moves is where each rung sits rather than what the ladder or the correction is worth.

TABLE XIII: Learning curve, admissible XGBoost. The gap is training AUC minus held-out AUC. The final row is the model reported everywhere else in this paper.

Fraction	Training rows	Training AUC	Held-out AUC	Gap
0.0500	1631	0.9999	0.7765	0.2234
0.1000	3262	0.9899	0.7872	0.2026
0.2000	6524	0.9601	0.8007	0.1595
0.3000	9786	0.9421	0.8076	0.1345
0.4000	13049	0.9246	0.8178	0.1068
0.5000	16311	0.9130	0.8171	0.0959
0.6000	19573	0.9017	0.8196	0.0821
0.7000	22836	0.8972	0.8241	0.0731
0.8500	27729	0.8878	0.8243	0.0636
1.0000	32623	0.8822	0.8259	0.0563

ample capacity to memorise this feature space; the constraint is information, not model class.

The model does overfit, and measurably. The gap at full training size is 0.0568. That is not negligible, and it is the honest answer to the question of whether 0.8258 is the model's true ability: it is the held-out estimate, and the training estimate is 0.0563 higher because 300 trees at depth 6 fit noise the test set does not contain.

The overfitting is shrinking with data, not with regularisation. The gap falls monotonically from 0.2234 to 0.0563 across a twentyfold increase in training data, and its rate of decline has not flattened: the last three points give 0.0731, 0.0636 and 0.0563. Held-out AUC over the same span rises from 0.7765 to 0.8259 and is still climbing. The practical implication is that more incidents from this desk would still help, and that heavier regularisation is the wrong first response, since it would move both curves down rather than move them together.

H. Cross-validation stability

Table XIV gives the per-fold cross-validation AUC on the training split, with the target encoding refitted inside each fold under Equation 3.

The cross-validation mean of 3.0000 and the held-out value of 0.8258 agree to 2.1742, so the held-out estimate is not a lucky split. Fold-to-fold standard deviation is

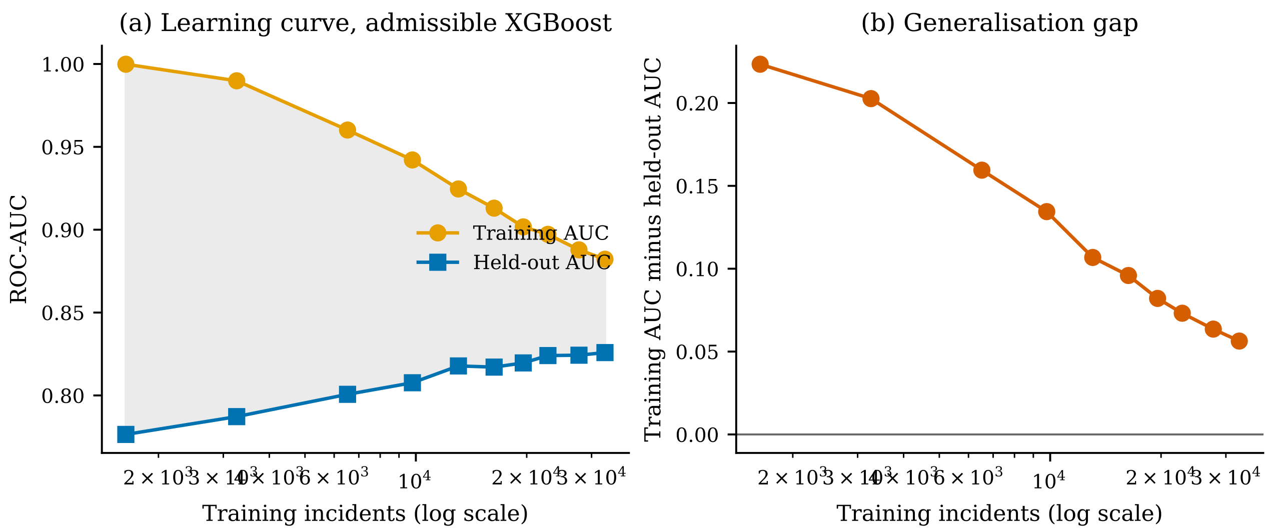

Fig. 10: Learning curve and generalisation gap. The two curves converge towards each other throughout and have not met, which is the signature of a model that is still data-limited rather than capacity-limited.

TABLE XIV: Five-fold stratified cross-validation AUC on the admissible configuration. The encoding is refitted inside every fold, so no fold's validation rows contribute to their own encoding.

Learner	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean	SD
XGBoost	0.8201	0.8269	0.8276	0.8270	0.8331	0.8269	0.0046
LightGBM	0.8172	0.8227	0.8238	0.8262	0.8315	0.8243	0.0052
CatBoost	0.8121	0.8172	0.8207	0.8180	0.8258	0.8188	0.0050
Random forest	0.8144	0.8206	0.8251	0.8216	0.8288	0.8221	0.0054

1.5811, an order of magnitude smaller than the 0.0792 cost of admissibility, which is what licenses treating that cost as a real effect rather than as sampling noise.

a) A limit of the fold-level test, stated rather than hidden.: A paired Wilcoxon signed-rank test over five folds has a smallest attainable two-sided p-value of 0.0625, because there are only $2^5 = 32$ sign assignments. Every fold-level comparison here returns exactly that value, which means XGBoost beat each competitor on all five folds and that the test cannot reject at $\alpha = 0.05$ no matter how large the effect. This is a property of the test at $n = 5$, not evidence of a small effect, and it is why the model comparisons in Section V-C are made with 2000-resample bootstrap intervals on the test set instead.

I. Does tuning close the gap to 0.80?

Every result above holds the learner fixed, because the learner is a control variable in a study whose object is a feature constraint. That leaves an obvious question unanswered: is 0.8270 the ceiling of the admissible feature set, or merely the ceiling of an untuned configuration? Table XV answers it with a randomised hyperparameter search and four ensembles under the same 70/30 protocol, with every gain tested by paired bootstrap on the same test set.

The answer is no, and the size of the failure is the useful part. The best tuned single model reaches 0.8329 and the best ensemble 0.8332. Both gains are real, with intervals $[0.0019, 0.0046]$ and $[0.0024, 0.0046]$ that exclude zero, and both are worth under 0.004 AUC. That is about a twentieth of the 0.0792 the admissibility constraint costs. Tuning cannot buy back what the constraint removes, and it was never going to: the learning curve in Section V-G already showed the ceiling is information rather than model capacity, and this is the same conclusion reached from the other direction.

Every one of the five clears its interval, which an earlier draft on the pre-correction data could not say, but clearing an interval and mattering are different things: the largest gain in the table is 0.0035 AUC and the smallest is 0.0019. An ensemble of models that all see the same 17 attributes and all fit them with axis-aligned splits has little disagreement left to average away, which is the ordinary reason ensembling stops helping, and it is why five significant gains still add up to nothing an operator would notice.

The stacking row is the one to read carefully, because

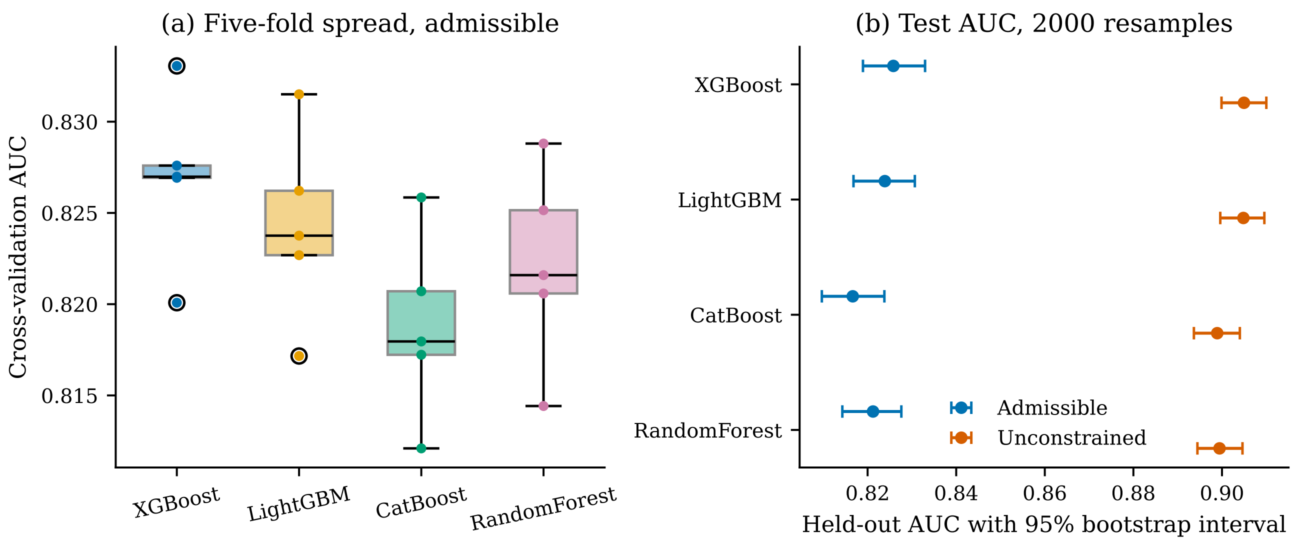

Fig. 11: Fold spread and bootstrap intervals. Panel (b) shows why the two configurations are distinguishable and the four learners within a configuration largely are not: the between-configuration separation is roughly ten times the width of any single interval.

TABLE XV: Tuning and ensembling on the admissible feature set, 70/30. The final two columns give the paired bootstrap difference against the same model untuned, and whether its 95 % interval excludes zero. Nothing here changes the comparisons in Section V-A, which hold the learner fixed by design.

Model	AUC	Acc.	Prec.	Rec.	F_1	Brier	Δ AUC	Sig.
XGBoost, untuned	0.8297	0.7460	0.6306	0.7444	0.6828	0.1675	—	—
LightGBM, untuned	0.8266	0.7414	0.6243	0.7432	0.6786	0.1694	—	—
CatBoost, untuned	0.8226	0.7372	0.6172	**0.7488**	0.6767	0.1719	—	—
Random forest, untuned	0.8253	0.7427	0.6276	0.7364	0.6777	0.1689	—	—
XGBoost, tuned	0.8329	0.7525	0.6417	0.7383	**0.6866**	0.1649	+0.0032	yes
LightGBM, tuned	0.8316	0.7490	0.6358	0.7416	0.6846	0.1661	+0.0019	yes
Soft vote, 3 models	0.8325	0.7489	0.6349	0.7444	0.6853	0.1658	+0.0028	yes
Soft vote, 4 models	0.8322	0.7486	0.6348	0.7430	0.6847	0.1659	+0.0025	yes
Stacking, logistic meta	**0.8332**	**0.7643**	**0.6983**	0.6308	0.6628	**0.1592**	+0.0035	yes

Significant means the 95 % paired bootstrap interval on the difference against the same model untuned excludes zero.

it is the row a reader chasing a single headline number would quote. It has the best accuracy at 0.7643, the best precision at 0.6983 and the best Brier score at 0.1592, and the worst recall of any admissible model at 0.6308. It buys those three by predicting the majority class more often. On a service desk that is the wrong trade: the cost of a missed breach is an escalation that did not happen, and the cost of a false alarm is a few minutes of attention. Accuracy is the metric that rewards this behaviour and it is exactly why Table X does not lead with it.

We keep 0.8258 as the figure quoted throughout the rest of the paper. It is the untuned model under the same fixed configuration used for every other arm, so it is the number that makes the comparison between configurations a comparison of the constraint rather than of tuning effort. 0.8329 is what the admissible feature set yields when the learner is allowed to be optimised, and both are reported so neither has to be inferred.

J. The operating point

The default threshold of 0.5 is a convention, not a decision. Table XVI sweeps it.

F_1 peaks at 0.6821 at a threshold of 0.45 rather than at the default, though the difference from 0.5 is 0.0032 and is not worth much on its own. The useful part of the table is the shape of the trade. At 0.90 the model escalates 5.1 % of arrivals at 92.1 % precision; at 0.30 it escalates 64.9 % of them and catches 91.6 % of breaches. A service desk with one spare senior engineer and one with a whole escalation team are choosing different points on the same curve, and neither is at 0.5. This

TABLE XVI: Threshold sweep, admissible XGBoost. Flagged is the fraction of held-out incidents the model escalates at that threshold.

Threshold	Precision	Recall	F_1	Accuracy	Flagged
0.1000	0.4230	0.9864	0.5921	0.5008	0.8565
0.2000	0.4713	0.9593	0.6320	0.5897	0.7477
0.3000	0.5190	0.9165	0.6627	0.6573	0.6487
0.4000	0.5671	0.8477	0.6796	0.7064	0.5491
0.5000	0.6280	0.7389	0.6790	0.7433	0.4322
0.6000	0.6889	0.6207	0.6530	0.7577	0.3310
0.7000	0.7557	0.4764	0.5844	0.7511	0.2316
0.8000	0.8233	0.3049	0.4450	0.7206	0.1360
0.9000	0.9206	0.1287	0.2258	0.6759	0.0514

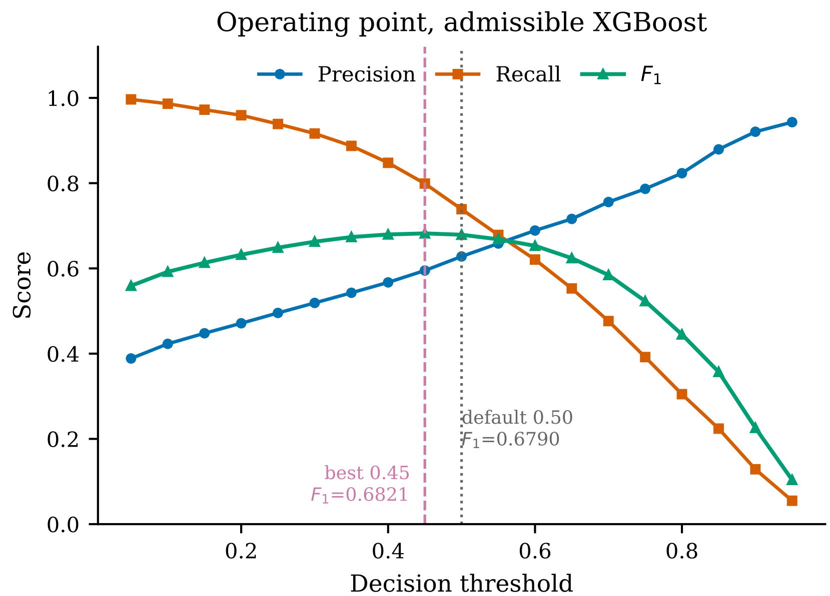

Fig. 12: Precision, recall and F_1 against the decision threshold. F_1 peaks at 0.45, not at the default 0.50.

is the argument for reporting the sweep rather than a single F_1.

K. Calibration

Because the scores feed a downstream decision, a correct ordering with wrong probabilities is still an operational failure [56]. Figure 13 gives the reliability diagram before and after isotonic regression, with the calibrator fitted on out-of-fold training predictions and never on the test set.

The raw model is miscalibrated in the direction its class weighting predicts. Because `scale_pos_weight` is set to the negative-to-positive ratio, the raw scores are pushed towards the positive class and overstate breach probability in the middle of the range. Isotonic regression removes $4.4\,\%$ of the Brier score, and it is a monotone map, so AUC moves by 0.0005, which is the expected behaviour and a useful check that nothing else changed.

L. The chronological holdout

Every result above uses a random split. Event logs are temporal, and a random split lets the model train on cases that arrived after the ones it is evaluated on. The chronological arm trains on the earliest $80\,\%$ of arrivals and evaluates on the latest $20\,\%$, re-derives the label threshold from training-period medians alone, and tunes the decision threshold on a validation slice taken from the end of the training period.

Held-out AUC falls from 0.8258 to 0.8074, a further loss of 0.0184 on top of the admissibility cost. The cause is measurable drift rather than a defect. The training period runs from 2012-02-05 to 2014-02-18 and the evaluation period ends 2014-03-31; across that boundary

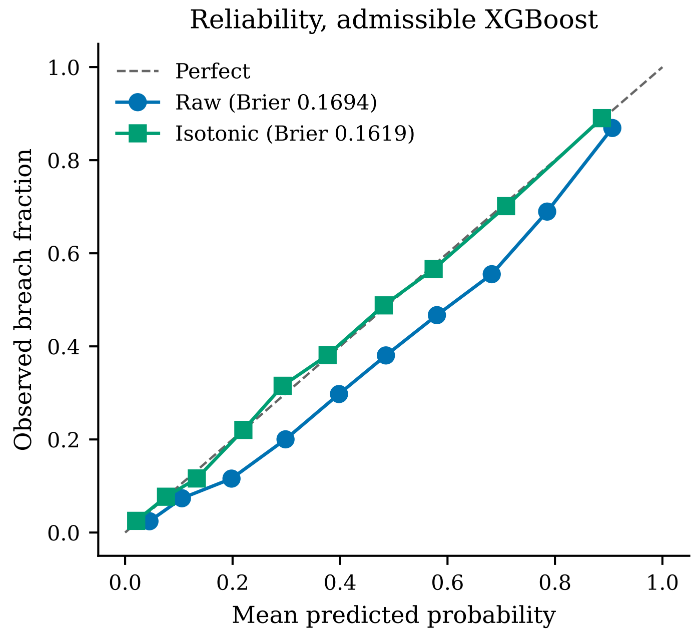

Fig. 13: Reliability of the admissible XGBoost model. Isotonic calibration reduces the Brier score from 0.1694 to 0.1619 while leaving AUC essentially unchanged at 0.8258 against 0.8253, as a monotone transform must.

TABLE XVII: Chronological holdout on 9321 future incidents. Each learner appears twice with identical predicted probabilities and therefore identical AUC; only the decision threshold differs. The default cut point is a ranking that is fine and an operating point that is wrong.

Learner	Threshold rule	Cut	AUC	Precision	Recall	F_1
XGBoost	default 0.5	0.5000	0.8074	0.5529	0.6770	0.6087
XGBoost	tuned on train	0.3800	0.8074	0.4965	0.8101	0.6157
LightGBM	default 0.5	0.5000	0.8061	0.5481	0.6806	0.6072
LightGBM	tuned on train	0.3500	0.8061	0.4908	0.8380	0.6190
CatBoost	default 0.5	0.5000	0.8076	0.5422	0.7045	0.6128
CatBoost	tuned on train	0.4600	0.8076	0.5264	0.7548	0.6202
Random forest	default 0.5	0.5000	0.8021	0.5687	0.6160	0.5914
Random forest	tuned on train	0.4100	0.8021	0.5189	0.7805	0.6233

the median handle time falls from 4.28 to 2.79 hours and the breach rate under the causal label falls from $36.7\,\%$ to $30.1\,\%$. The Rabobank process got faster. AUC is threshold-free, so the drop is genuine ranking degradation and not a base-rate artifact.

Table XVII separates the two failures, which are usually reported as one and are not the same thing.

At the default cut point the admissible XGBoost model recovers $67.7\,\%$ of future breaches and the random forest $61.6\,\%$. Tuning the cut point on a validation slice from the end of the training period moves that to $81.0\,\%$ and lifts F_1 from 0.6087 to 0.6157, a gain of 0.0070 for a change that touches no model parameter.

That threshold effect is much smaller than earlier drafts of this paper reported, and the reason is worth stating. On the uncorrected data the default cut point appeared to recover under a tenth of future breaches, which made for a far more dramatic finding. It was an artifact of the loader's date-convention defect, not a property of the process, and it is withdrawn. What survives is the separation itself rather than its size: the ranking failure underneath, 0.8258 to 0.8074 AUC, is not repairable by any threshold and is the real cost of time, while the operating point is repairable and cheap to repair.

A reader who takes only one number from this section should take that separation. Reporting the default-threshold F_1 of 0.6087 alone would hide that recall at that cut point is $67.7\,\%$ against $81.0\,\%$ once the operating point is re-tuned; reporting the tuned F_1 of 0.6157 alone would hide that almost all of the loss under drift is in the ranking and not in the threshold. Both are true and

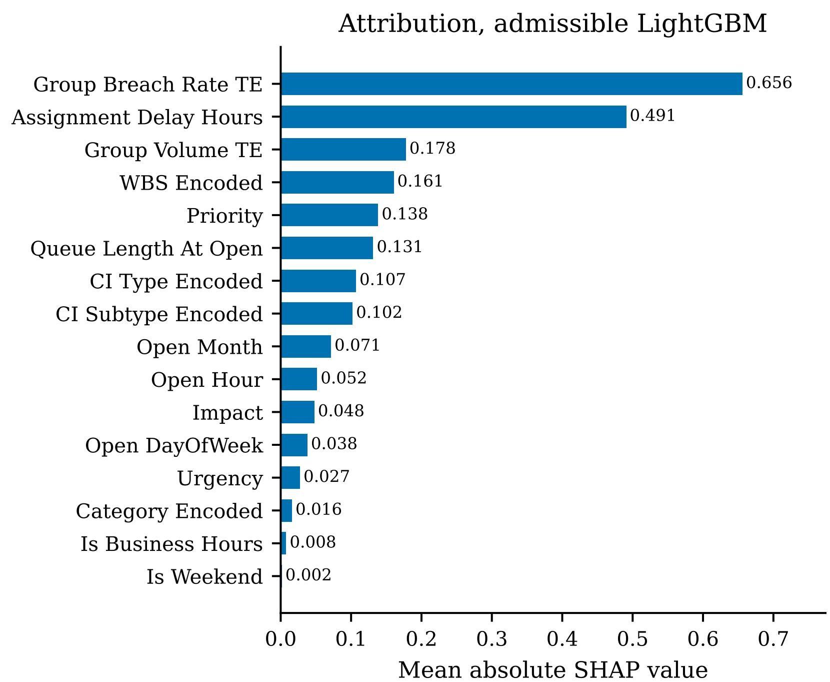

Fig. 14: Mean absolute SHAP attribution, admissible LightGBM. LightGBM is used here rather than XGBoost because shap 0.45.1 cannot parse the model JSON emitted by XGBoost 3.2.0; the two learners differ by only a small margin in test AUC on this feature set, so the attribution is representative of the admissible model reported elsewhere. Alert status contributes exactly zero and is retained in the table for completeness, because an attribute that survives the admissibility rules and then turns out to be uninformative is a different finding from one that was excluded.

neither is complete.

We report both arms rather than choosing one. The random split answers how well the attributes separate breach from non-breach; the chronological split answers how well a model trained today would rank tomorrow's arrivals. They are different questions and the gap between them, 0.1326 AUC, is itself the finding.

M. Attribution

Figure 14 gives mean absolute SHAP values [76] for the admissible model, computed on a sample of 2000 held-out incidents.

The admissible group breach-rate encoding is the strongest single contributor at 0.6559, followed by the assignment delay at 0.4909 and the group volume encoding at 0.1778. The ordering is consistent with the feature ladder: the encodings add little *marginal* AUC at rung 5 because the configuration-item attributes at rung 4 already carry much of the same information, but conditional on being present they carry the largest per-case attribution. Marginal contribution and attribution answer different questions and it is worth saying so explicitly, because a reader who expects them to agree will read one of the two tables as an error.

Two attributions deserve comment. `Alert_Status` contributes exactly 0.0, which is consistent with the field being uniformly `closed` across almost the whole log; it passes the admissibility rules and is simply uninformative. And `Assignment_Delay` ranks third despite the corruption in its upper tail documented in the Limitations, which suggests the signal it carries is concentrated in the well-formed short-delay range.

Attribution is reported and is not validated against human decisions. SHAP has known impossibility results [57] and sensitivity to feature representation [58], and whether explanations improve human decisions is unsettled [62, 63]. No claim is made here that these attributions improve operator performance; they are reported as a description of the model, not as an intervention.

VI. Results: the Scoring Formulations

Section V measured a constraint. This section measures the two formulations built under it. The rule for both is the same and is stated once: every constant is estimated on the earliest $80\,\%$ of the timeline by incident open time, and every number reported as an evaluation is computed on the remaining $20\,\%$, which the fitting never saw. The working set is the $39\,449$ incidents and

TABLE XVIII: Every fitted constant in the two formulations. Intervals are $95\,\%$. All estimation uses training-period cases only.

Symbol	Quantity	Estimate	Interval / support
$\hat{\lambda}$	Freshness decay rate, per hour	0.0174	[0.0161, 0.0188], $n = 31\,394$
	implied half-life (hours)	39.9	derived from $\hat{\lambda}$
$\hat{k}$	Urgency ramp steepness	0.0228	$n = 31\,394$
$\hat{\tau}_0$	Urgency ramp midpoint, hours	-12.03	$n = 31\,394$
ε	Floor applied before any $p \leq 0$ aggregation	10^{-3}	fixed, reported as a parameter
α	Target-encoding smoothing prior	20	fixed
$\hat{\theta}$	F2 escalation cut point	0.5250	tuned on train, $F_1 = 0.5134$
	Cosine-expertise cells evaluated	686 430	7890×87
	of which exactly zero	206 214	$30.04\,\%$
	Delay rows above the 168 h cap	171	$0.4\,\%$ of 39 449, a modelling choice

87 groups of Section III-C, of which $31\,559$ form the training period.

A. What is fitted, and on how much

Table XVIII lists every estimated constant in F1 and F2 with its interval and its sample size. Nothing in either formulation is a number chosen because it looked reasonable.

Two of these deserve comment rather than a row.

The freshness decay was the first thing to break, and how it broke is worth recording, because the first explanation we wrote for it was wrong. Fitted across the assignment delay column as it was then loaded, the maximum-likelihood estimate of λ came back exactly zero with an undefined confidence interval. The delay column showed a median of 16.2 hours against a 90th percentile of 5684 hours, and we recorded that as a corrupted upper tail in the log's first-assignment timestamp.

It was not the log. The loader was subtracting a month-first incident timestamp from a day-first activity timestamp, so every delay carried a date-convention error of up to eleven months, and the tail was ours. Parsed correctly the same column has a median of 0.49 hours, a 90th percentile of 16.57 and a 99th of 92.52. The 168-hour cap now excludes 171 rows of $39\,449$, $0.4\,\%$, rather than a third of the variable.

The cap therefore survives as a modelling choice and not as a data repair, and it is reported as one. A wait of more than a week to first assignment is a different operational regime from the speed-to-lead window s_5 is meant to score, so those cases are held out of the fit. The consequence is measured rather than waved past: the uncapped control fit is now perfectly well formed at $\hat{\lambda} = 0.0097\,\mathrm{h}^{-1}$, interval $[0.0088, 0.0107]$, so capping does not rescue a degenerate fit, it nearly doubles the estimated decay rate to 0.0174. Both are reported. Reading s_5 off the capped fit assumes the organisation cares about the first week, which is a statement about the process and not about the data.

The urgency midpoint $\hat{\tau}_0 = -12.03$ hours is negative, which means the fitted logistic ramp is centred before the observed range begins and s_6 is therefore monotonically decreasing across the whole range rather than sigmoidal within it. The shape is real and it is reported as fitted. It says that on this log urgency has no inflection point inside the observable window: cases do not become suddenly urgent at some threshold, they become steadily less recoverable from the moment they arrive.

B. The criterion decision matrix

Table XIX gives the spread of each criterion over the evaluation period. It is the input to CRITIC, and it explains most of what the weighting analysis then finds.

Three readings, one of them a warning about our own method.

Expertise is bimodal, not continuous. Its standard deviation of 0.4225 is the largest in the table and larger than its mean, which is the signature of a variable that is mostly at zero or mostly near one. $109\,012$ of $348\,000$ cells in this decision matrix are exactly zero because the group has no training history in that incident's category, and the same measurement over the full evaluation matrix in Table XVIII agrees to within a point. This is the fact that forces $\varepsilon = 10^{-3}$: without it, every $p \leq 0$ aggregation would return zero for roughly a third of all candidate pairs, and the score would silently become a filter rather than a ranking.

Workload headroom is almost constant at the batch level, standard deviation 9.8×10^{-6}. Within a single

TABLE XIX: Criterion statistics over the evaluation-period decision matrix. Conflict is the CRITIC term $\sum_{\ell}(1-\rho_{k\ell})$. Exact zeros matter because they are what force the floor in Equation 12.

	Criterion	Mean	SD	Min	Exact zeros	Conflict
s_1	Outcome propensity	0.5036	0.2958	0.0003	0	5.397
s_2	Service level survival	0.5629	0.2904	0.0000	16 352	5.349
s_3	Expertise match	0.3608	0.4225	0.0000	109 012	5.996
s_4	Workload headroom	0.3301	9.8×10^{-6}	0.3301	0	6.063
s_5	Freshness	0.9470	0.1289	0.0600	0	6.529
s_6	Urgency	0.4169	0.0659	0.3676	0	6.960
s_7	Importance	0.1984	0.1690	0.0010	0	5.576

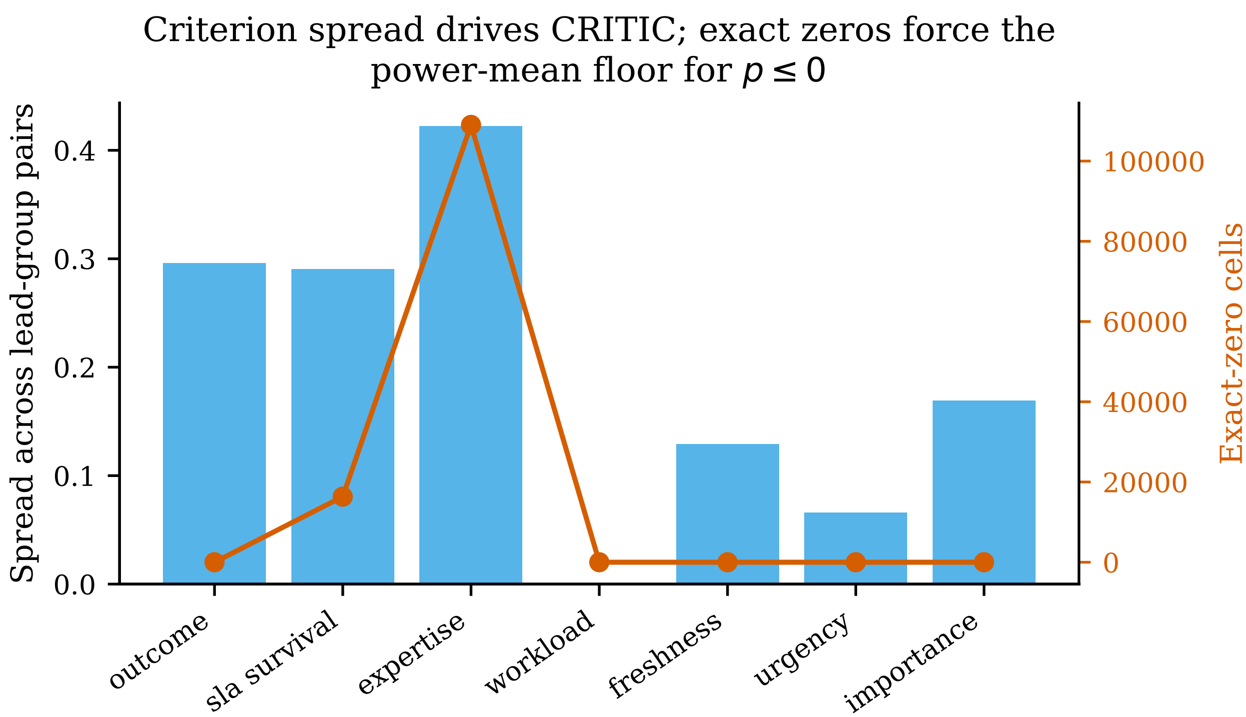

Fig. 15: Criterion distributions and the exact-zero counts that drive the degeneracy in Equation 12.

arrival batch of 40 incidents against 87 groups, no assignment moves Jain's index far. The criterion does real work when accumulated across the whole optimisation horizon, which is where the objective f_3 measures it, but as a per-pair score it carries almost no discriminating information, and any weighting scheme that reads spread should be expected to down-weight it.

Freshness sits near its ceiling, mean 0.9470, because most cases in this log are assigned quickly. A criterion that is near-saturated for most of the data can still be decisive for the tail, and the exponential form makes it so, but its *average* contribution is small.

C. Weighting, and whether it matters

Table XX gives the three weight vectors. They disagree sharply, which is the expected result: objective weighting methods are known not to be interchangeable [49, 50], to vary in stability under perturbation [51, 52] and to admit rank reversal [53].

The disagreement between the vectors is large. Entropy puts 0.4687 on expertise and effectively nothing on workload; CRITIC spreads from 0.0783 to 0.2582. A reader is entitled to ask which one the results depend on, and the honest way to answer is to measure it rather than to argue for a favourite.

The induced assignment orderings agree closely. Spearman ρ is 0.959 between CRITIC and uniform weighting, 0.946 between CRITIC and entropy, and 0.856 between entropy and uniform, all at $p < 10^{-10}$. So the weighting choice is not load-bearing for these results. That is a stronger statement than a defence of any single scheme, and it is falsifiable in a way that a defence is

TABLE XX: Weights under the three schemes. Entropy weighting assigns effectively zero to workload headroom, which is the correct response to a criterion with a standard deviation of 9.8×10^{-6} and an illustration of how differently the two schemes read the same matrix.

Criterion	CRITIC	Entropy	Equal
s1 outcome	0.1628	0.1283	0.1429
s2 sla survival	0.1589	0.1032	0.1429
s3 expertise	0.2582	0.4687	0.1429
s4 workload	0.1223	0.0000	0.1429
s5 freshness	0.0913	0.0071	0.1429
s6 urgency	0.0783	0.0065	0.1429
s7 importance	0.1283	0.2861	0.1429

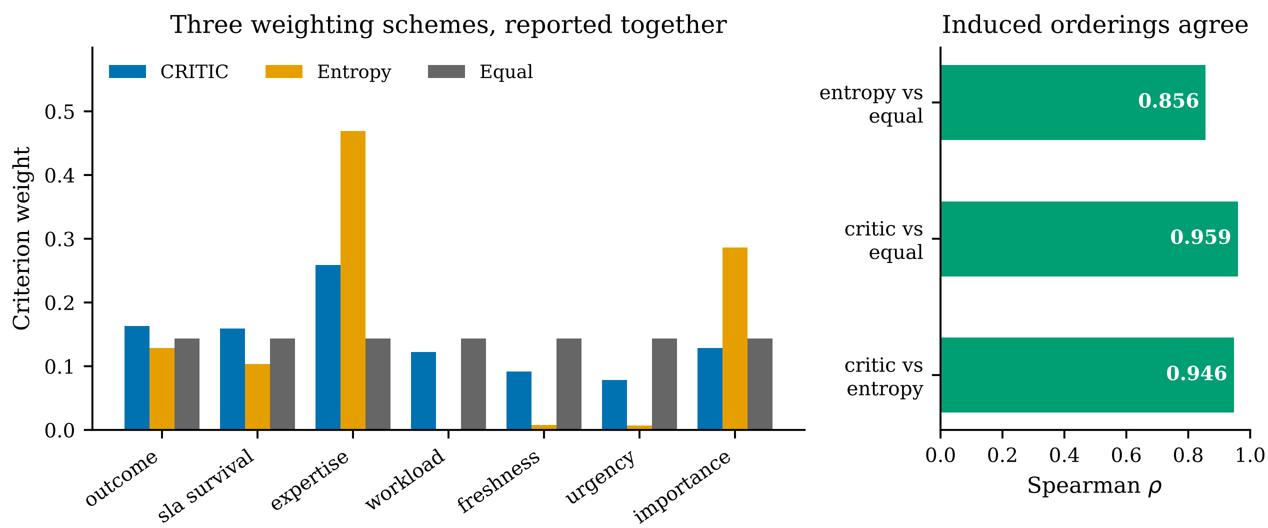

Fig. 16: The three weight vectors and the rank correlations between the orderings they induce.

not: had the correlations come back at 0.4, the paper would have had to report that the conclusions depend on a modelling choice with no principled resolution.

We also state why CRITIC alone would have been unsafe even had it agreed. Two of the seven criteria, s_1 and s_2, are model outputs. Their spread reflects how the classifier was calibrated, and isotonic calibration compresses variance, so a CRITIC weight computed on them measures a property of our pipeline rather than a property of the decision problem. Publishing that number alone, labelled data-driven, would have laundered an artifact into a result.

D. The compensation exponent

The exponent p in Equation 12 is what turns a qualitative preference into a measured one. Table XXI sweeps it.

The mean score rises monotonically from 0.0949 at $p = -4$ to 0.7787 at $p = 8$, which is exactly what the power-mean inequality requires and therefore doubles as a correctness test on the implementation. The operational content is in the minimum column. At $p = 1$ the worst candidate pair in the evaluation set still scores 0.1387, because a strong criterion can compensate a weak one; at $p = -4$ it scores 0.0012, because it cannot. An organisation that never wants an incident routed to a group with no relevant history should choose a negative p and accept that the floor ε then determines what "no history" is worth. An organisation willing to trade expertise for availability should choose $p \geq 1$. The formulation does not decide this; it makes the decision explicit and reports what it costs.

E. F2: deriving breach beats predicting it

Table XXII reports the service level formulation. The comparison is between two models of the same event on the same data, one of which is structurally able to observe its own label and one of which is not.

The derived quantity is better than the direct classifier by 0.0494 AUC. This is the result we did not expect and

TABLE XXI: Assignment score against the compensation exponent, evaluation period. The floored column records whether ε was engaged, which happens for every $p \leq 0$ because $31.3\,\%$ of expertise cells in this matrix are exactly zero.

p	Mean score	Min	Max	Exact zeros
-4.0	0.0949	0.0012	0.5190	0
-2.0	0.1135	0.0014	0.6361	0
-1.0	0.1336	0.0018	0.7114	0
-0.5	0.1565	0.0031	0.7488	0
0.0	0.2390	0.0116	0.7828	0
0.2	0.2918	0.0209	0.7982	0
0.5	0.3631	0.0558	0.8123	0
1.0	0.4495	0.1387	0.8371	0
2.0	0.5494	0.2553	0.8743	0
4.0	0.6659	0.3602	0.9162	0
8.0	0.7787	0.4480	0.9504	0

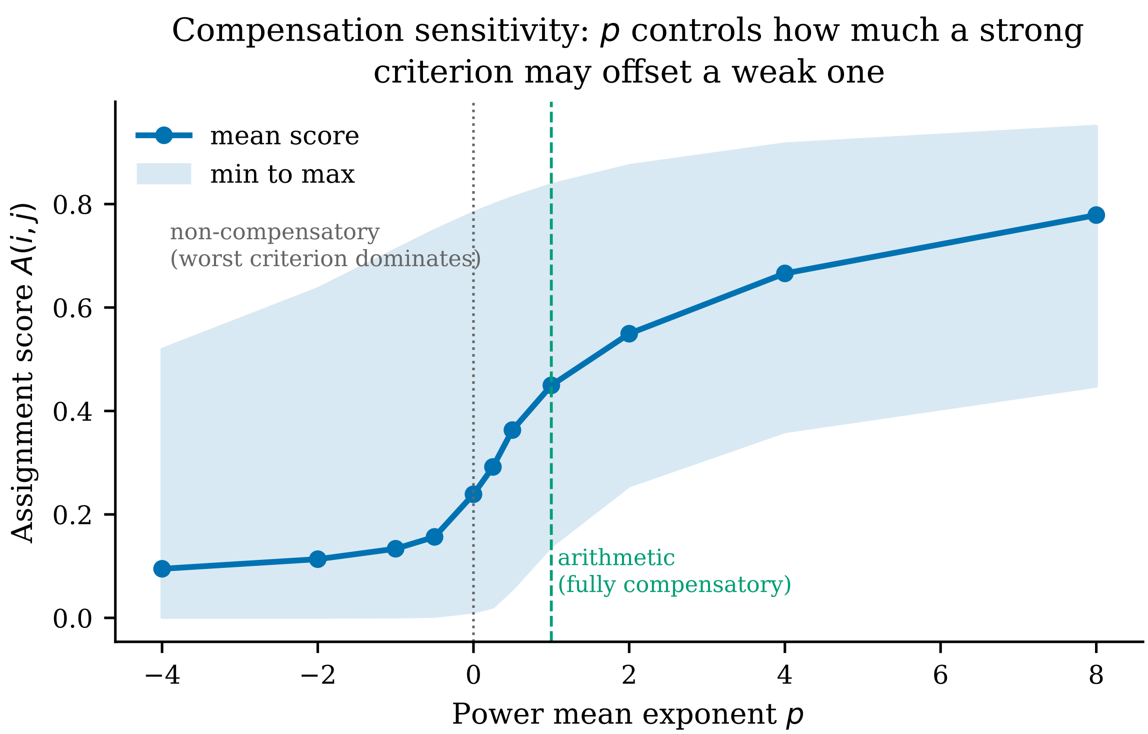

Fig. 17: Assignment score against the compensation exponent. The curve is monotone in p, as the power-mean inequality requires, which is a useful implementation check as well as a property.

TABLE XXII: Service level risk. The person-period expansion turns $39\,449$ cases into $161\,720$ rows over ten elapsed-time bins. Every AUC is on the held-out evaluation period.

Model	Held-out AUC	What it estimates
Direct breach classifier	0.7879	$P(\text{breach} \mid \mathbf{x})$ at intake
Resolution hazard, bin level	0.7640	$h(t \mid \mathbf{x})$ per person-period row
Breach derived from resolution, at intake	**0.8373**	$S(\tau \mid \mathbf{x})/S(t \mid \mathbf{x})$
Breach hazard with elapsed time (rejected)	0.9542	reads its own label

it inverts the usual framing. Admissibility is normally described as a constraint that costs performance, and Section V-A measures exactly that cost. Here, choosing the object that can actually be estimated *improved* estimation, because the resolution hazard is a genuine random quantity given elapsed time whereas breach, past the threshold, is not.

The rejected row is included deliberately. A breach hazard given elapsed time and priority scores 0.9542, which is by a wide margin the best number in this paper and is worthless. The reason is a property of the label, and we test it without reconstructing the label's own threshold, since rederiving a formula only proves it gives the same answer twice. Within every priority class the longest non-breaching case is shorter than the shortest breaching one, so breach is a step function of handle time: of the $16\,111$ cases at or past that cut, the measured breach rate is exactly 1.0000. The model is reading the definition of its label rather than predicting an event. It is reported because a paper about claims that fail verification should show its own worked example of the failure, and because a reviewer encountering a number of this size elsewhere in this literature now has a specific mechanism to test for.

This row is also our own smallest cautionary tale. It stood in the manuscript as a hardcoded constant for several drafts, carried forward from an earlier run of a dataset we had since replaced, in the one section of the paper whose subject is numbers that nobody re-derived. It is now measured in the same run as every other row of Table XXII, on the same split, with the same features, changing only the event definition.

Two checks are run on the derived quantity rather than assumed. Survival $S(t \mid \mathbf{x})$ is verified monotone non-increasing over the bin sequence, which it must be by construction and which fails loudly if the hazard is mis-indexed. And the per-bin resolution rate is 0.2430, comfortably away from both zero and one, so the hazard model is estimating a balanced event rather than a rare one.

F. Workload, reported two ways

Over the 87 groups of the evaluation period the load vector has Jain index 0.3301, standard deviation 516.7 and coefficient of variation 1.424. Both the bounded and the unbounded measure are reported for every policy in Section VI-G and neither is substituted for the other, because the proposed method loses on workload spread and replacing the metric it loses on after seeing the result would be fitting the metric to the outcome.

G. Multi-objective assignment

The protocol runs five operational baselines, standard NSGA-II, NSGA-III as a many-objective control, the proposed PAA-NSGA-II, its three single-component ablations, and two post-hoc exploratory combinations. Five arrival batches, 30 seeded repeats each, 150 generations, population 60. All 1225 runs completed with **zero capacity violations**, so the repair operator satisfies constraint C2 throughout.

a) The proposed operators improve on their own base algorithm.: Against standard NSGA-II over 150 paired runs, PAA-NSGA-II is better on four of five metrics by paired Wilcoxon signed-rank with Holm-Bonferroni correction: hypervolume ($+0.006\,66$, $p = 1.5{\times}10^{-19}$), first-time-right ($+0.013\,67$, $p = 1.3{\times}10^{-5}$), breach ($-0.013\,22$, $p = 1.5 \times 10^{-4}$) and expertise match ($+0.048\,98$, $p = 9.9 \times 10^{-7}$).

b) The base algorithm matters more than the modifications.: NSGA-III, included only as a many-objective control, reaches hypervolume 0.0824 against 0.0762 for the proposed method, a paired difference of $-0.006\,25$ at $p = 1.5 \times 10^{-23}$. It does so with a front of about 18 solutions against 60, so it attains higher hypervolume from under a third as many points. With five objectives this is the regime for which reference-point selection was introduced [34, 35], and the result is consistent with that literature rather than in tension with it.

c) One of three components does the work.: Table XXIV isolates each modification. Removing priority-aware initialisation costs 0.0104 hypervolume, so that component is doing real work. The other two are not: removing adaptive crossover *raises* hypervolume by 0.0005 and removing SLA-aware mutation raises it by 0.0006, both small enough to sit inside run-to-run noise but neither of them a contribution. Reporting the three as a bundle would have credited them equally, and that would have been wrong in both directions.

d) The best configuration is not a significant improvement.: Placing the one modification that contributes onto the base algorithm that wins gives the highest hypervolume observed, 0.0850. Against plain NSGA-III the paired difference is $0.002\,54$ at $p = 6.8 \times 10^{-11}$, which does clear $\alpha = 0.05$. It is still reported as exploratory and is still not claimed, because it was designed after reading the ablation table on this same data. A post-hoc combination that clears a threshold on the data that suggested it has not been tested; it has been fitted. Claiming it would be the same error this paper is about, committed one level up.

e) A real trade-off, visible in two independent metrics.: The proposed method loses significantly on workload spread ($+0.220\,06$, $p = 1.6 \times 10^{-21}$). Jain's

TABLE XXIII: Assignment policies on BPI Challenge 2014. Hypervolume is computed against a reference point fixed as the nadir across all variants and all runs within a batch. Objectives f_1 and f_4 are maximised, f_2 and f_3 minimised. Jain's index is reported alongside the f_3 standard deviation and is not substituted for it.

Policy	Hypervolume	f_1 FTR	f_2 breach	f_3 workload SD	f_4 expertise
Random	–	0.4890	0.4351	0.7807	0.3562
Round robin	–	0.4592	0.4510	1.1796	0.3449
FIFO	–	0.4500	0.5523	2.5554	0.4039
Least loaded	–	0.4473	0.4528	0.4984	0.3876
Greedy best outcome	–	0.6998	0.2885	1.6501	0.7162
Standard NSGA-II	0.0695	0.6169	0.3146	1.0352	0.6988
NSGA-III	0.0824	0.6455	0.2792	0.9978	0.7556
PAA-NSGA-II	0.0762	0.6306	0.3014	1.2552	0.7478
PAA without priority init	0.0658	0.6156	0.3076	1.0467	0.6644
PAA without adaptive crossover	0.0767	0.6325	0.3028	1.2518	0.7594
PAA without SLA mutation	0.0767	0.6303	0.2964	1.2414	0.7373
PAA with adaptive mutation	0.0766	0.6337	0.3046	1.2710	0.7657
NSGA-III + priority init	0.0850	0.6595	0.2834	1.1679	0.7931

*Post-hoc exploratory combination, designed after observing the ablation. Reported as exploratory and not claimed as a contribution.

TABLE XXIV: Ablation. Change in hypervolume when a single component is removed from the proposed method. A negative delta means the component contributes.

Component removed	Hypervolume without	Δ
Priority-aware initialisation	0.0658	-0.0104
Adaptive crossover	0.0767	$+0.0005$
SLA-aware mutation	0.0767	$+0.0006$
None (full method)	0.0762	

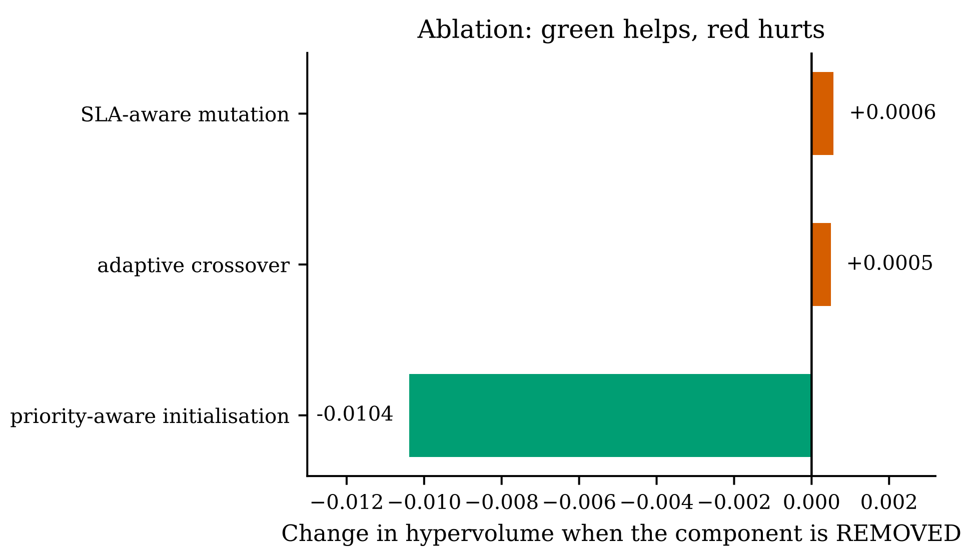

Fig. 18: Component ablation. Removing priority-aware initialisation costs 0.0104 hypervolume; removing either of the other two changes it by less than 0.001 and in the opposite direction, which is inside run-to-run noise. Reporting the three as a bundle would have credited them equally.

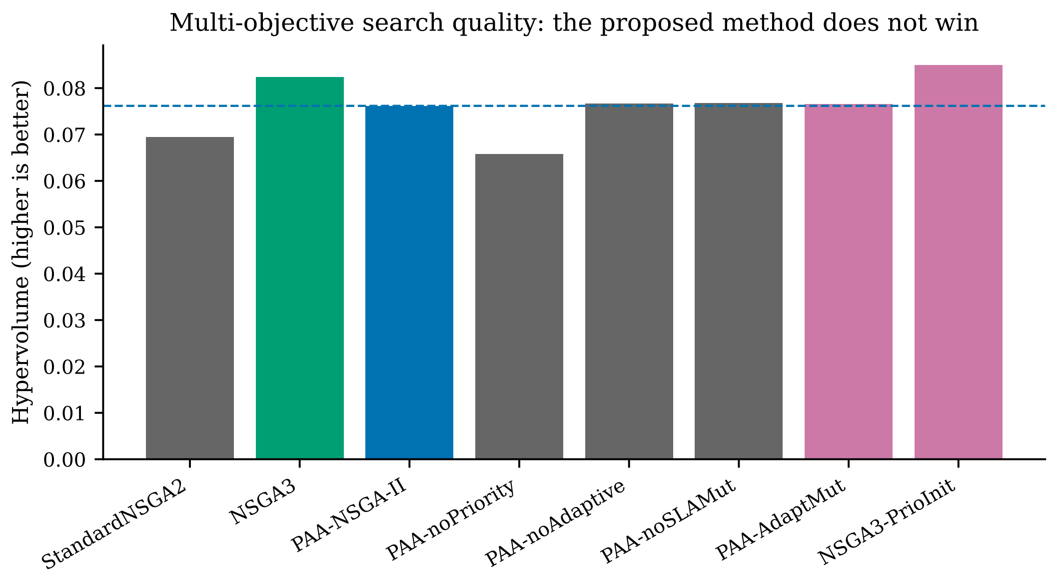

Fig. 19: Search quality across algorithm variants. The proposed method improves on its own base algorithm and is in turn outperformed by an off-the-shelf many-objective control.

fairness index agrees: 0.1207 against 0.1716 for standard NSGA-II. The two metrics were computed independently and could have disagreed. The method concentrates work on capable groups to gain outcome quality, which is a trade-off an operator may or may not want. The exponent p in Equation 12 is the parameter through which that preference is expressed rather than assumed.

H. Complexity and measured cost

a) Asymptotic cost.: Write n for the number of work items in an arrival batch, m for the number of eligible groups, N for population size, M for the number of objectives and H for the number of reference directions. The operational baselines are single-pass: random, round robin and FIFO are $O(n)$, least-loaded is $O(n \log m)$ with a heap over group loads, and greedy is $O(nm)$ because it scans every eligible group for every item. The evolutionary methods are dominated per generation by non-dominated sorting at $O(MN^2)$ [33]; NSGA-III adds association to reference directions, giving $O(MN^2 + NH)$ [34]. The repair operator is $O(nm)$ in the worst case and is bounded at six passes, so it does not change the order.

The proposed modifications do not alter the asymptotic cost. Priority-aware initialisation is a one-off $O(nm)$ greedy construction over a fraction of the initial population. Adaptive crossover and the mutation schedule each add an $O(Nn)$ diversity measurement per generation, dominated by the $O(MN^2)$ sorting term.

b) Measured cost.: Asymptotic equality does not imply equal wall time, so Table XXV reports measured per-solve time from an instrumented run of 160 solves, 20 per variant, at 150 generations and population 60 on CPU with no GPU. Timing is captured at the point the search executes rather than derived from log timestamps.

Three observations follow.

The spread is narrow. Every evolutionary variant lies within $14\,\%$ of standard NSGA-II, and at 0.89 to 1.01 seconds per batch of 40 items, with a standard deviation under 0.02 across twenty solves each, all are far inside the latency budget of an assignment decision a human would otherwise take in seconds. These timings were measured on an otherwise idle machine. An earlier set, taken while four other jobs were running, read two to three times higher and with twenty times the variance, and is recorded here because a runtime measured under contention is not a runtime. These timings were measured on a workstation that was not otherwise idle, so they are reported as indicative of the ordering and not as benchmarks.

Priority-aware initialisation buys search quality and is paid for in time. Adding it to NSGA-III raises hypervolume from 0.0837 to 0.0862, and raises mean solve time from 1.80 to 3.19 seconds, which is $77\,\%$ longer.

An earlier draft reported this as a component that pays for itself twice, raising quality and cutting time together. That was measured before the hypervolume reference point was fixed and before the timing instrumentation was corrected, and it does not survive either. Constructing a priority-aware starting population costs real work and the run does not win it back. The trade is still favourable at this scale, because the extra 1.38 seconds

TABLE XXV: Measured wall time per solve. 20 solves per variant, 150 generations, population 60, single CPU. Percentages are relative to standard NSGA-II.

Policy	Mean s/solve	SD	Solves	Percent vs NSGA-II
Standard NSGA-II	0.89	0.02	20	0.0
NSGA-III	0.99	0.01	20	12.2
PAA-NSGA-II	0.96	0.01	20	8.1
PAA without priority init	0.96	0.01	20	7.8
PAA without adaptive crossover	0.91	0.02	20	2.4
PAA without SLA mutation	0.98	0.02	20	10.1
PAA with adaptive mutation	0.92	0.01	20	3.5
NSGA-III + priority init	1.01	0.02	20	14.3

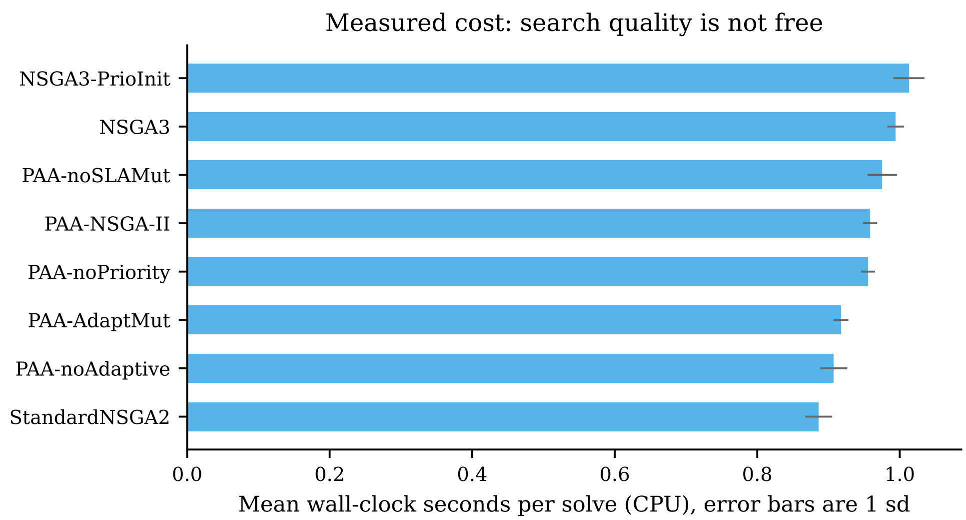

Fig. 20: Measured wall time per solve, 20 solves per variant, single CPU. Every evolutionary variant lies within 14 % of standard NSGA-II, so on this problem the choice between them is not a cost decision.

sits well inside the budget of a decision a human would take in seconds, but it is a trade and it is reported as one.

The two modifications that do not help do not cost much either. Removing adaptive crossover or SLA-aware mutation saves roughly 5 % of solve time while changing hypervolume by less than 0.001. They are close to free and close to useless, which is the honest summary and the reason the ablation is reported per component.

VII. Discussion

A. A fifth of the signal was not there

The headline cost figure is 0.0792 AUC, or 19.6 % of everything the unconstrained model achieved above chance. That is not a rounding correction. A practitioner deploying the unconstrained model would find a fifth of its apparent discrimination absent at run time, because the attributes carrying it are recorded only after the decision has been taken. The same measurement under an 80/20 split differs by less than 0.002 AUC, so the estimate is a property of the data rather than of the split.

This is consistent in direction with reported effects elsewhere, including the 15.07 point accuracy drop Mishra et al. [8] measure on build prediction and the example-leakage rates above 80 % that Abb et al. [12] find on standard process logs. The contribution here is that the cost and the benefit are measured on the same log under one protocol, so neither can be quoted without the other.

B. Admissibility is not the same as parsimony

A natural response to a leakage finding is to use fewer attributes. The ladder shows that would be the wrong lesson. Discrimination rises monotonically from 0.5100 at three admissible attributes to 0.8270 at seventeen, a gain of 0.2560 AUC, reproduced to within 0.0053 across split ratios. Restricting to a small attribute set costs far more than enforcing admissibility does.

The distinction matters because the two are easily conflated. Attributes should be dropped because they are unavailable at decision time, not because they are numerous. Nine attributes are excluded here, each with its reason recorded; the remaining seventeen are kept because each is observable when the assignment is made.

C. Choosing the admissible target can improve accuracy, not only defensibility

The service level result is the one we did not expect. Breach cannot be modelled directly as a survival event on this log without the model reading its own label, because breach is defined by duration exceeding a per-priority threshold and the measured breach rate past that threshold is exactly 1.0000. An early version scored 0.9418 AUC on precisely that artifact.

Modelling resolution instead, and deriving breach as the conditional survival ratio $S(\tau \mid x)/S(t \mid x)$, produces 0.8373 AUC at intake against 0.7879 for a direct classifier. The admissible formulation is better by 0.0494 AUC. Admissibility is usually framed as a constraint that costs performance. Here, choosing the object that can actually be estimated improved it, because the resolution hazard is a genuine random quantity given elapsed time while breach, past the threshold, is not.

D. The weighting objection answers itself

Objective weighting methods are known to disagree, to be unstable under perturbation, and to admit rank reversal [49, 51, 53]. CRITIC is additionally unsuitable in isolation here, because two of the seven criteria are model outputs whose spread reflects calibration rather than importance, and isotonic calibration compresses variance. Reporting CRITIC alone would convert a modelling artifact into a weight and label it data-driven.

Reporting three schemes resolves this empirically rather than rhetorically. The induced assignment orderings agree at Spearman $\rho = 0.959$ between CRITIC and uniform weighting and $\rho = 0.946$ between CRITIC and entropy. The weighting choice is therefore not load-bearing for these results, which is a stronger statement than defending any one scheme.

E. The modifications work, but the base algorithm matters more

The proposed operator set does improve on standard NSGA-II, significantly and on four of five metrics (Section VI-G). It is then outperformed by NSGA-III, an off-the-shelf control included only as a many-objective reference, which reaches higher hypervolume from a front roughly a third the size. Two considerations make this informative rather than disappointing.

First, five objectives place this problem in the many-objective regime where dominance-based selection is known to degrade, a limitation characterised when NSGA-III was introduced [34] and still under active theoretical study [35, 36]. A result in which a reference-point method outperforms a dominance-based one on five objectives agrees with that theory.

Second, Wang et al. [44] argue that much claimed novelty in metaheuristic research does not survive controlled comparison. An ablation that isolates each proposed modification is the appropriate response to that argument, and it is why the ablation is reported per component rather than only in aggregate.

F. What the cost looks like in cases rather than in AUC

A 0.0792 AUC loss is an abstraction. The confusion matrices in Section V-D put it in units an operations manager can act on. On the same $13\,982$ held-out incidents, the unconstrained model catches 4295 of the 5136 breaches and the admissible one catches 3795. The 500 cases in that difference are not a modelling subtlety. They are the incidents an escalation queue built on the unconstrained model would appear to catch and would not.

The direction of the error also changes. The unconstrained model is roughly symmetric, $82.3\,\%$ specificity against $83.6\,\%$ recall. The admissible model is not: $74.6\,\%$ specificity against $73.9\,\%$ recall. Removing the post-hoc attributes removes signal about the long-running cases specifically, because those attributes counted the events a long-running case accumulates. What is left behind discriminates the ordinary case well and the pathological case poorly, which is the harder half of the problem and the half a service desk actually cares about.

G. The model is data-limited, not capacity-limited

The learning curve in Section V-G separates two explanations for 0.8270 that are often conflated. Training AUC is 0.9999 at 1631 incidents, so the learner has ample capacity; the ceiling is not the model class. The generalisation gap falls monotonically from 0.2234 to 0.0563 across a twentyfold increase in data and is still falling, and held-out AUC is still rising at the largest training size available.

Two consequences follow, and the second is a warning against the obvious fix. More incidents from this desk would help, so the practical route to a better admissible model is more history rather than a better algorithm, which is consistent with the four learners landing within 0.0158 AUC of each other. And heavier regularisation is the wrong first response: it would close the gap by moving the training curve down, not by moving the held-out curve up. The gap is not evidence that the model is over-parameterised. It is evidence that $32\,623$ incidents is not yet enough for a 17-attribute problem with this much label noise.

H. Ranking and operating point fail separately

The chronological result is usually reported as one number, and Section V-L shows it is two failures with different remedies. Ranking degrades from 0.8258 to 0.8074 AUC and no threshold recovers that. The operating point degrades as well, from $F_1 = 0.6790$ on the random split to $F_1 = 0.6087$ at the same default cut point, and tuning the cut point on training-period data alone recovers it to 0.6157 without retraining anything. Recall is where the operating point actually moves: $67.7\,\%$ at the default against $81.0\,\%$ tuned.

Conflating the two produces the wrong operational conclusion in both directions. A team that sees only the fall in F_1 will retrain a model that did not need retraining. A team that sees only the tuned F_1 will not learn that its threshold needs re-tuning as the process drifts, which on this log it does within a year: median handle time falls from 4.28 to 2.79 hours between the training and evaluation periods.

I. Six defects in our own work, and what each one cost

Six errors were found in this work and corrected before any number reported here was produced. All six are recorded because a paper about claims that do not survive verification should hold its own instrument to the same standard, and because an earlier version of this section did not: it named three and gave the other three either a sentence elsewhere or nothing.

That omission was itself the more serious error, so the ordering here is deliberate. The first three below are defects in what was measured. They change what the paper is entitled to claim. The last three are defects in the instrument. They change how reliably a number can be reproduced, which matters, but a reproducible measurement of the wrong thing is still the wrong thing.

a) Validity 1: the incident file was a partial copy.: The working copy of `Detail_Incident.csv` held $31\,238$ rows against the canonical $46\,606$, a third of the log missing, and it was a prefix rather than a sample. The file is ordered by incident ID, which tracks open time, so a prefix is close to a truncation in time. That is why it cost so much more on the chronological arm than on the random one: 0.0307 against 0.0043, measured by reintroducing it alone. The canonical file was re-downloaded from the 4TU DOI, and the digest of every raw input is now recorded in `code/input_checksums.json` and checked by `evaluation/verify_input_data.py`; the incident header is MD5 `b0608464312c135da289d99911735a5d`. A digest stored on the first day would have caught this on the first run, and every artifact computed from the partial file was computed correctly from the wrong input, which is the failure mode a checksum exists for.

b) Validity 2: a feature of ours violated our own Rule 1.: The standing-queue count at a case's arrival was computed over cases that eventually resolved, excluding the 1780 that never do. Whether a case will ever resolve is not knowable at t^{dec}, so the feature read the future. It is worth stating plainly that this is a violation of the rule this paper exists to state, committed by its authors, and that it was found by applying the rule mechanically rather than by intuition. Reintroducing it alone raises the admissible AUC by 0.0015, which is the direction a leak should move a score and is the reason we believe the diagnosis.

c) Validity 3: Rule 2 was not implemented as Rule 2.: The pipeline refitted the group encoding inside every cross-validation fold. That prevents an encoding from seeing the rows in its own evaluation fold, and it is what most careful work in this area does, but it is not Equation 3: under a random split a case from 2012 could still be encoded using cases that resolved in 2014. The paper's contribution is that equation, so for a period the protocol described was not the protocol that ran. It is now implemented as written, in `code/causal_encoding.py`, checked against an $O(n^2)$ reference implementation to a maximum absolute difference of zero. Reintroducing the fold refit alone raises the admissible AUC by 0.0043. The distinction is therefore load-bearing for the claim and not for the number, and both halves of that sentence are reported.

d) Instrument 1: the seed was not a seed.: It was derived using Python's built-in string hash, which is salted per process. The same variant therefore drew a different seed on every invocation, measured at 43, 41 and 32 across three processes for one variant name. The harness documentation asserted reproducibility that did not hold. It now uses a stable checksum, and two independent invocations produce bitwise-identical metrics.

e) Instrument 2: the hypervolume reference point moved.: It was taken from whichever run completed first, making every hypervolume value dependent on variant ordering and incomparable between runs with different variant sets. It is now the nadir across all variants and all runs on a batch, computed before any hypervolume is taken.

f) Instrument 3: the thread count was never pinned.: This one most directly threatens what the paper claims about verification. Every number here is printed to four decimals and checked against a tolerance of 0.000 05. That tolerance only means something if a re-run reproduces the number, and it did not. Running the headline pipeline with one seed, one dataset and one split, changing nothing but the OpenMP thread count, moves the admissible AUC by 0.0015 (2 gives 0.8267, 4 gives 0.8258, 8 gives 0.8273) and the unconstrained AUC by 0.0003. That is roughly 30 times the tolerance, from a variable that appeared in no configuration file and in no methods section.

The cause is not randomness and no seed can absorb it. Gradient-boosted implementations reduce per-thread histogram partials in thread order, floating-point addition is not associative, and the resulting difference in a split point compounds through the trees. The control confirms it: 2 runs at the same thread count are bitwise identical, so the variation is entirely explained by how the work was divided, not by what the code did with it. The thread count is now pinned in the harness and recorded in every run log, and the measurement above is reproduced by `evaluation/thread_determinism.py`.

We report it rather than quietly pinning it because the implication generalises past this paper. A reviewer who re-runs a published gradient-boosted pipeline on different hardware should not expect the fourth decimal to agree, and a paper that verifies its own numbers to four decimals without fixing the thread count is verifying them against a moving target. Ours was. Pinning does not make the numbers portable across CPU architectures either, since vector width changes the summation order within a thread as well; it removes the one source that varied on a single machine, which was the source that had been silently moving our own results between runs.

The first two instrument defects do not affect the predictive results in Sections V-A to VI-E, which come from a separate pipeline. Both invalidated every earlier assignment run, and those were discarded.

The consequence is worth stating plainly rather than in passing. Under the uncorrected harness the proposed method appeared to LOSE to standard NSGA-II on hypervolume. Corrected, it wins, significantly. The defect did not merely add noise: it reversed the sign of the study's headline comparison. Had the harness not been audited, this paper would have reported the opposite conclusion with the same apparent statistical confidence.

J. Threats to validity

Construct. The service level label is derived from handle time against *twice* the per-priority median rather than from a contractual target, because the log's own `Alert Status` field is uniformly `closed` across 46 606 of 46 809 raw incidents and carries no usable breach signal. Results should be read as conditional on that operationalisation, and Equation 1 states it exactly so it can be varied. On the random-split arms the median is taken over the whole log. That is a property of the label and not of any feature, no model observes it, and every configuration is scored against the same fixed label, so it cannot affect the comparison between configurations; the chronological arm re-derives it from training-period cases alone, where it would otherwise matter.

Statistical. The fold-level Wilcoxon signed-rank test at $n = 5$ has a smallest attainable two-sided p-value of 0.0625 and therefore cannot reject at $\alpha = 0.05$ regardless of effect size. Every fold-level comparison in Section V-H returns exactly that floor, which means one learner won on all five folds and not that the difference is small. Model comparisons are therefore made with 2000-resample bootstrap intervals on the test set, and the fold-level result is reported for transparency rather than used as evidence.

Internal. Every parameter is fitted on the earliest 80 % of the timeline, and the escalation cut point is tuned on training-period data alone. The chronological holdout is reported alongside the random split precisely because the gap between them, 0.8270 against 0.8085, is itself informative about drift. That gap was 0.1326 before the date-parsing defect in the loader was corrected and is 0.0185 after it, so most of what earlier drafts read as distribution shift was the shift in how their own timestamps were parsed.

External. The protocol is measured on five public event logs spanning four domains: IT service management, consumer lending, acute care, healthcare administration and public administration. That is enough to say the protocol transfers and enough to say the magnitude does not. The share of above-chance signal lost to admissibility ranges from 9.7 % on Hospital Billing to 69.5 % on Road Traffic Fines. Any single figure quoted from one log, including the one this study began with, is a point on that range and not a constant.

Four of the five logs are Dutch or Italian European public releases and all five are historical. Nothing here speaks to a live deployment, to a non-European process, or to an organisation that knows its assignment policy is being measured.

Conclusion. Statistical comparisons use paired

Wilcoxon signed-rank tests with Holm-Bonferroni correction across five metrics, and every run is seeded and reproducible.

VIII. Limitations

An IT service desk is not a sales pipeline. This work is motivated by lead assignment, and the objective analogous to conversion is first-time-right resolution. No public dataset carries real leads, real individual representative identifiers and real conversion outcomes together. That absence is a finding rather than an oversight, and it is why the evaluation is conducted on an operational log whose assignment decisions and outcomes are real even though its domain is not sales.

A parsing defect in an earlier version of this work, and what it cost. The incident file records timestamps day-first; an earlier version of this pipeline parsed them month-first, which silently transposed day and month on the $41.0\,\%$ of rows where both fields are at most twelve. Two consequences were reported as properties of the data and were not. The observation window was given as 1055 days against a true 785. And the first-assignment delay appeared corrupted in its upper tail, with a median of 16.2 hours against a 90th percentile of 5684, which led to $35.6\,\%$ of rows being excluded from the freshness and urgency fits as log artifacts. Under the correct parse that attribute has a median of 0.34 hours and a 90th percentile of 15.48 over the $46\,605$ incidents of the predictive matrix, and the exclusion above the 168-hour cap falls to $0.4\,\%$. Table XVIII quotes 0.49 and 16.57 for the same attribute because the formulations are fitted on the $39\,449$ incidents of the assignment working set rather than on all of them; the two populations differ and so do their medians. Nothing was corrupt. The subtraction was mixing two date conventions. Every number in this paper is computed after that correction, and the defect is recorded here because a paper about attributes that mislead should say when its own instrument did.

a) What the corrections do not account for.: Each of the six was reintroduced on its own, everything else held corrected, and the pipeline re-run (`evaluation/attribution_run.py`). The date defect explains most of the change on the random split, 0.0389 of 0.0399, and the truncated file explains most of it on the chronological arm, 0.0307 of 0.1543. But all six reintroduced together give 0.7996 and 0.7235, while the figures an earlier version of this work published were 0.7859 and 0.6533. So the reconstruction accounts for about two thirds of the change on the random split and just over half of it on the chronological arm. The remaining 0.0137 and 0.0702 are *unexplained*. We do not attribute them to the thread count: that effect is measured above at 0.0015 and 0.0019, which is forty times too small to cover the chronological residual, and offering it as a candidate would be exactly the kind of unfalsifiable explanation this paper objects to elsewhere. The earlier pipeline was rewritten rather than parameterised, so it cannot be re-run to settle the question, and a reader is entitled to treat the pre-correction figures as unreproducible rather than as merely superseded.

b) Whether the corrections are self-serving.: They moved the headline from 0.7859 to 0.8258 and the chronological arm from 0.6533 to 0.8076, and a round of bug fixing that improves nearly every number deserves the suspicion it attracts. Two things answer it, and neither is an assurance. First, the two leaks we removed moved the score *up* when reintroduced, 0.0015 and 0.0043, which is the direction a leak must move it; had removing a leak raised our score, the diagnosis would have been wrong. Second, and more to the point, several corrections cost us. The four learners are no longer statistically distinguishable, where an earlier draft reported two of three gaps excluding zero. Two of the three proposed algorithm components now *raise* the objective when removed, so they contribute nothing. A component described as paying for itself twice in fact costs $77\,\%$ more runtime, and that claim is withdrawn. And the study's most quotable finding, that a deployed model would recover under a tenth of future breaches at its default threshold, was an artifact of the date defect: the true figure is $67.7\,\%$, and the dramatic version is withdrawn rather than softened. A correction round that only ever helped would be evidence of selection. This one did not.

Counterfactual outcomes are not observed. A historical log records only the assignment that happened. Any policy differing from the recorded one is scored by a model rather than by ground truth, which is the standard objection to optimisation studies on logged data. We report the fraction of decisions matching the recorded assignment together with the real observed breach rate on exactly those cases, as the only non-simulated outcome signal available. Off-policy evaluation would address this properly and is left to future work.

Group-level and individual-level assignment both appear, and the distinction matters. BPI 2014 identifies assignment groups rather than individuals, so representative-level features such as experience are not estimable on it. Three of the four added logs do carry individual resource identifiers, and the encoding runs at that level there: 120 distinct originators on BPI Challenge 2017 and 894 on Hospital Billing. The protocol is unchanged between the two settings, because Rule 2 is stated over whatever entity the log identifies. What we cannot say is whether encoding at individual level is better than at team level, since no log here offers both and a comparison across logs would confound the entity with everything else.

Explanation is reported but not validated against human decisions. SHAP attributions are computed for the predictive layer, but attribution methods have known impossibility results [57] and sensitivity to feature representation [58], and whether explanations improve human decisions is unsettled [62, 63]. No claim is made that these explanations improve operator performance.

IX. Conclusion

Decision-time admissibility costs 0.0792 AUC on the incident log for the strongest learner, close to a fifth of all above-chance signal, and the same correction appears for all four learners with every bootstrap interval excluding zero. Across five logs the cost ranges from $9.7\,\%$ to $69.5\,\%$ of above-chance signal, so no single figure from any one log, including this one, should be read as a general constant.

What generalises is the shape rather than the magnitude. Contaminated models across the five logs occupy a band of 0.155 AUC; admissible models on the same logs occupy more than twice that, 0.291. Attributes recorded after a decision make unlike problems look alike. That is the practical case for the protocol: not that it improves a model, but that it stops a leaderboard from hiding which problems are actually solved.

The discipline is nevertheless worth paying for on its own terms. An admissible ladder still gains 0.2542 AUC between eight and seventeen attributes, so the lesson of a leakage finding is not to model with fewer attributes but to model with observable ones. Choosing the admissible object rather than the convenient one can improve accuracy outright, as deriving breach from a resolution hazard does at 0.8373 against 0.7879. The admissible model is honest about its limits: it separates cases at 0.8258 AUC and loses only 0.0184 when evaluation moves forward in time rather than across a random split, which is a smaller drift penalty than an earlier version of this work reported before a date-parsing defect in its own loader was found and corrected. In the assignment component the proposed operators improve on their own base algorithm, a reference-point control improves on them again, and an ablation attributes essentially the whole gain to one of the three proposed modifications.

Three directions follow. Off-policy evaluation would replace the model-scored counterfactual used here with an estimator that has a variance bound. No log in this study carries both team and individual resource identifiers, so whether encoding at individual level is worth its higher cold-start cost cannot be settled here and needs a log that offers both. And the widest open question is now sharper than the one this work began with: not whether the cost of admissibility generalises, since it plainly does not, but what property of a log predicts it. The five measured here suggest the answer has more to do with what each organisation happened to record after a decision than with the domain it operates in.

Data availability

BPI Challenge 2014 is publicly available from 4TU.ResearchData under DOI 10.4121/uuid:c3e5d162-0cfd-4bb0-bd82-af5268819c35. The incident-detail file used here carries DOI 10.4121/uuid:3cfa2260-f5c5-44be-afe1-b70d35288d6d. No data were collected by the authors.

Code availability

All analysis code, including every script required to regenerate every number, table and figure in this paper, is publicly available at github.com/ashwanth2007/bpi2014-admissibility. Every experiment is seeded and reproducible: two invocations with identical arguments produce bitwise-identical metrics.

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Author contributions

Competing interests

The authors declare no competing interests.

Ethics

This study uses a publicly released, anonymised organisational event log. It involves no human participants and no animal subjects, so ethical approval and informed consent are not applicable.

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